

University of Heidelberg
Faculty of Mathematics and Computer Science
Institute of Computer Science
Research Group: Data Analysis and Modeling in Medicine

Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

Name: Youssef Daoudi El Boukhrissi
Student ID: 4277022
Supervisor: Prof. Dr. Jürgen Hesser
Submission date: 28 February 2026

Abstract

Contents

1. Introduction	1
1.1. Research Questions	1
1.2. Structure of the Thesis	2
2. Theoretical Background	3
2.1. Pinhole Camera Model	3
2.2. Lens Distortion	4
2.2.1. Radial Distortion	4
2.2.2. Tangential Distortion	5
2.3. Camera Calibration	6
2.3.1. Target-Based Camera Calibration	6
2.3.2. Calibration Parameters and Error Metrics	7
2.3.3. Bundle Adjustment	8
2.4. Light Field Camera Systems	9
2.4.1. Multi-Camera Light Field Systems	9
2.4.2. Micro-Lens-Array-Based Light Field Cameras	10
2.5. State of the Art in Light Field Camera Calibration	10
3. Methods and Materials	12
3.1. Methods	12
3.1.1. Overall approach	12
3.1.2. Use Case Description	12
3.1.3. Requirements Analysis	14
3.1.4. UML Class Diagram	16
3.1.5. Health Monitoring and Decision Metrics	17
3.1.6. State Diagram	18
3.2. Materials	19
3.2.1. Runtime Environment and Toolchain	19
3.2.2. Synthetic Data Generation with Blender	19
3.2.3. Image Sets and Ground Truth Definition	21
4. Results and Evaluation	22
4.1. Results of Initial Calibration	22
4.1.1. Intrinsic Accuracy	22
4.1.2. Extrinsic Accuracy	23
4.1.3. Reprojection and RMS Error	24
4.1.4. Comparison of Mean Results Across Image Sets	25
4.2. Results of LiFCal-Based Recalibration	25
4.2.1. Reprojection Error after LiFCal	26
4.2.2. Health Score Comparison	27
5. Discussion of the Results	28
5.1. Interpretation of Initial Calibration Results	28
5.2. Limitations of LiFCal in the Given Setup	29
5.3. Comparison between Initial Calibration and LiFCal	30
5.4. Comparison to the State of the Art	30
5.5. Potential Extensions and Improvements	31

6. Conclusion and Outlook	33
6.1. Conclusion	33
6.2. Outlook	33
A. Appendix	35
A.1. Additional Results	35
A.1.1. Imageset1	35
A.1.2. Imageset2	37
A.1.3. Imageset3	39
A.1.4. Imageset4	41
A.1.5. Imageset6	43
A.1.6. Imageset7	45
A.2. Blender Scenes	47
Glossary	48

List of Figures

1.	Illustration of the pinhole camera model. Source: [19].	3
2.	Comparison of different types of radial distortion. Source: [21].	4
3.	Visualization of tangential distortion. Source: [19], [18].	5
4.	Example of a checkerboard and detected corner points. Source: [14]. . . .	6
5.	Visualization of reprojection error. Source: [2].	7
6.	Visual representation of bundle adjustment. Source: [15].	8
7.	Representation of a multi-camera system. Source: [29].	9
8.	Structure of a micro-lens-array-based light field camera. Source: [8]. . . .	10
9.	UML Class Diagram	16
10.	State Diagram	18
11.	Multi-camera array setup in Blender.	19
12.	Example of a Blender scene for the initial calibration with a checkerboard.	20
13.	Example of a Blender scene for LiFCal recalibration without a calibration target.	20
14.	Deviations of intrinsic parameters from ground truth for the initial calibration.	22
15.	Translation parameter deviations from ground truth and the initial calibration.	23
16.	Extrinsic parameter errors of the initial calibration.	23
17.	Reprojection error per camera of the initial calibration.	24
18.	Stereo RMS error of the initial calibration.	24
19.	Reprojection error per camera for a single LiFCal-based recalibration run.	26
20.	Reprojection error per camera across three LiFCal-based recalibration runs.	26
21.	Comparison of mean reprojection error per camera between initial calibration and LiFCal-based recalibration.	27
22.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset1).	35
23.	Extrinsic parameter errors of the initial calibration (Imageset1).	35
24.	Translation parameter deviations from ground truth and the initial calibration (Imageset1).	36
25.	Stereo RMS error of the initial calibration (Imageset1).	36
26.	Reprojection error per camera of the initial calibration (Imageset1). . . .	36
27.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset2).	37
28.	Extrinsic parameter errors of the initial calibration (Imageset2).	37
29.	Translation parameter deviations from ground truth and the initial calibration (Imageset2).	38
30.	Stereo RMS error of the initial calibration (Imageset2).	38
31.	Reprojection error per camera of the initial calibration (Imageset2). . . .	38
32.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset3).	39
33.	Extrinsic parameter errors of the initial calibration (Imageset3).	39
34.	Translation parameter deviations from ground truth and the initial calibration (imageset3).	40
35.	Stereo RMS error of the initial calibration (Imageset3).	40
36.	Reprojection error per camera of the initial calibration (Imageset3). . . .	40

37.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset4).	41
38.	Extrinsic parameter errors of the initial calibration (Imageset4).	41
39.	Translation parameter deviations from ground truth and the initial calibration (Imageset4).	42
40.	Stereo RMS error of the initial calibration (Imageset4).	42
41.	Reprojection error per camera of the initial calibration (Imageset4). . . .	42
42.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset6).	43
43.	Extrinsic parameter errors of the initial calibration (Imageset6).	43
44.	Translation parameter deviations from ground truth and the initial calibration (Imageset6).	44
45.	Stereo RMS error of the initial calibration (Imageset6).	44
46.	Reprojection error per camera of the initial calibration (Imageset6). . . .	44
47.	Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset7).	45
48.	Extrinsic parameter errors of the initial calibration (Imageset7).	45
49.	Translation parameter deviations from ground truth and the initial calibration (Imageset7).	46
50.	Stereo RMS error of the initial calibration (Imageset7).	46
51.	Reprojection error per camera of the initial calibration (Imageset7). . . .	46
52.	Blender scene for initial target-based calibration with checkerboard. . . .	47
53.	Blender scene for LiFCal-based recalibration without checkerboard. . . .	47

List of Tables

1.	Mean calibration results for different image sets	25
2.	Scenario-based summary of calibration results	25
3.	Overview of calibration runs and corresponding health scores	27

1. Introduction

In recent years, light field cameras have gained significance in many applications, such as robotics, augmented reality, and 3D reconstruction. In these application areas, the accuracy of the camera system plays a crucial role for overall performance. Camera calibration is therefore an essential step, as it enables an accurate geometric description of the imaging process and ensures reliable results.

There are various types of light field camera systems; however, this thesis focuses on multi-camera light field systems. These systems consist of multiple individual cameras arranged in a specific geometric configuration in order to capture different viewpoints of a scene. Compared to single-camera setups, the calibration of such systems is considerably more complex. The standard approach relies on target-based calibration using physical patterns such as checkerboards. In contrast, a more recent target-free calibration approach has been proposed by Fleith et al.[8]. The motivation of this thesis is to develop a calibration software that combines both approaches within a data-driven recalibration framework, enabling automatic calibration of a multi-camera array without human intervention.

Despite its potential, the proposed software approach also introduces limitations and challenges that must be carefully examined. For this reason, this thesis investigates the feasibility and performance of the target-free calibration method in the context of multi-camera light field systems. The research questions guiding this work are presented in the following section.

1.1. Research Questions

In order to systematically investigate the feasibility and performance of the target-free calibration method LiFCal for multi-camera light field systems, the following research questions are formulated. These questions aim to explore the applicability, accuracy, and conditions under which LiFCal can effectively replace traditional target-based calibration methods.

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

The first research question examines whether the LiFCal approach is suitable for the intended multi-camera light field system. The second research question investigates whether LiFCal can achieve a calibration quality comparable to a classical target-based approach and thus potentially replace it in the context of a multi-camera system. The third research question focuses on identifying the conditions and requirements necessary to enable a reliable and practical target-free recalibration within an automated software framework.

1.2. Structure of the Thesis

To provide the necessary background and context, the thesis begins with the theoretical foundations in Chapter 2. This chapter introduces the basic knowledge of camera models, lens distortion, camera calibration, and the different types of light field cameras. Chapter 3 describes the methods and materials used to implement the calibration framework. This includes the main idea of the project, the use case description, and the software architecture. The results of the implementation and evaluation are presented in Chapter 4. It contains the visualization of the results and their accuracy. Finally, Chapter 5 discusses the findings, answers the research questions, and provides an outlook on future work. A final conclusion completes the thesis in the last chapter.

2. Theoretical Background

This chapter introduced the theoretical foundations required for this thesis. It covered the pinhole camera model, including intrinsic and extrinsic parameters as well as lens distortion, as the basis for geometric camera modeling. Building on this, common camera calibration methods and error metrics were presented, including target-based calibration and bundle adjustment. The chapter also introduced light field camera systems, their main system types, and the state of the art in their calibration. Together, these concepts provide the necessary background for the calibration approach and experiments presented in the following chapters.

2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [17].

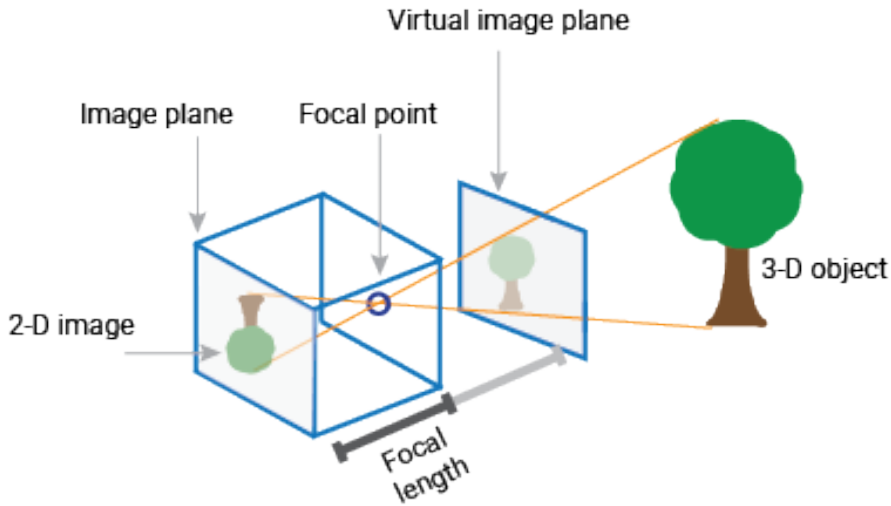


Figure 1: Illustration of the pinhole camera model. Source: [19].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [17]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [19].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where (f_x, f_y) are the focal lengths in pixel units along the x and y axes, s is the skew coefficient between the axes, and (c_x, c_y) is the principal point (optical center) in pixel coordinates. K is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

where R is a 3×3 rotation matrix representing the orientation of the camera with respect to the world coordinate system, and t is a 3×1 translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [19].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [7]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [23].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

2.2.1. Radial Distortion

Radial distortion is caused by the spherical shape of camera lenses, which leads to different refraction of light from the image center to the outer regions. This causes straight lines in the image to appear curved, resulting in barrel or pincushion distortion [6].

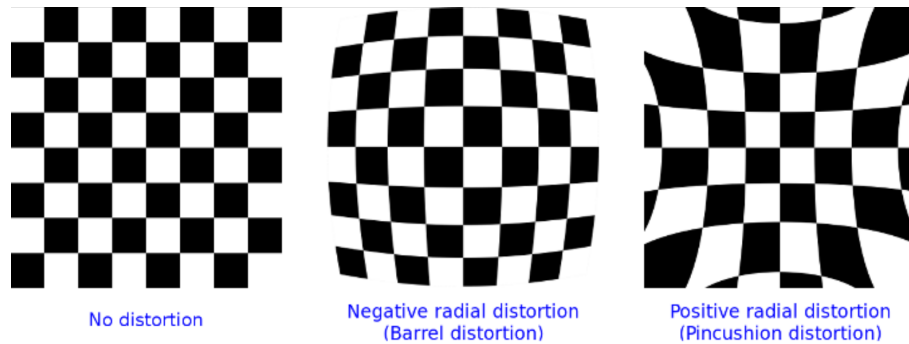


Figure 2: Comparison of different types of radial distortion. Source: [21].

The formal definition of radial distortion is given as follows [19]:

$$x_{distorted} = x \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (1)$$

$$y_{distorted} = y \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (2)$$

where x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (3)$$

The coefficients k_1 , k_2 , and k_3 in (1) and (2) control the strength of radial lens distortion. In practice, the first coefficient k_1 has the dominant influence on the distortion type: negative values of k_1 typically result in barrel distortion, while positive values lead to pincushion distortion.

2.2.2. Tangential Distortion

Tangential distortion occurs when the image sensor and the camera lens are not perfectly aligned or parallel to each other [19]. As a result, image points are displaced tangentially with respect to the optical axis. Figure 3 illustrates the geometric cause of tangential distortion and its effect on the captured image.

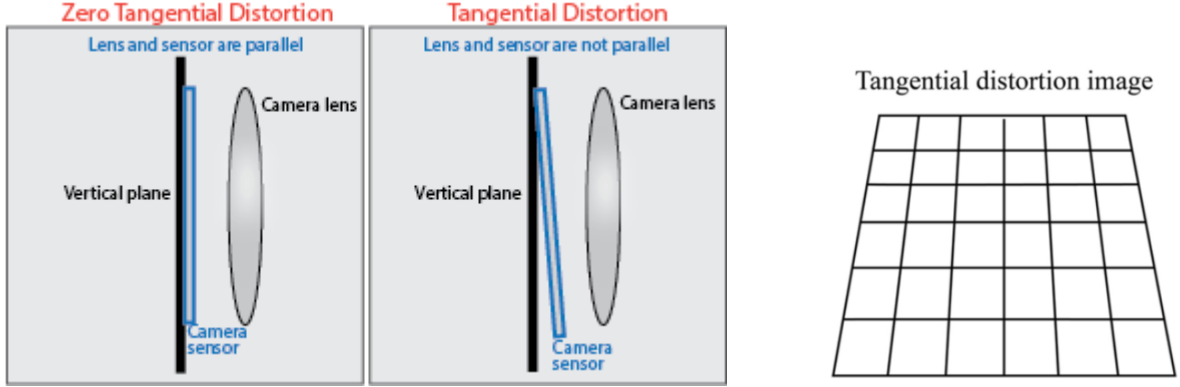


Figure 3: Visualization of tangential distortion. Source: [19], [18].

The formal definition of tangential distortion is given by the following equations [19]:

$$x_{distorted} = x + \left(2p_1 xy + p_2 (r^2 + 2x^2) \right) \quad (4)$$

$$y_{distorted} = y + \left(p_1 (r^2 + 2y^2) + 2p_2 xy \right) \quad (5)$$

Here, x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (6)$$

The coefficients p_1 and p_2 describe how strong the tangential lens distortion is. This type of distortion is caused by mechanical misalignment of the lens elements relative to the image sensor, resulting in a displacement of image points. Although tangential distortion is typically less pronounced than radial distortion, it can still negatively affect the camera calibration accuracy if not properly modeled.

2.3. Camera Calibration

The goal of camera calibration, also known as camera resectioning, is to estimate the parameters of a camera model. These parameters include the intrinsic parameters, the extrinsic parameters, and the lens distortion coefficients [19]. Knowing these parameters makes it possible to correct distortions in the recorded images and to perform metric measurements in the scene, for example to estimate object sizes or distances from image data.

Camera calibration is required for many tasks in 3D computer vision and photometry. Applications such as depth estimation, 3D reconstruction, or pose estimation rely on a precise mapping from 3D world coordinates to 2D image coordinates [25]. If the calibration is inaccurate, errors can propagate through the processing pipeline and reduce the quality of the final results.

In practice, cameras are often calibrated using targets with known geometric properties. Planar targets, such as checkerboard patterns, are widely used because they are easy to handle and lead to reliable results [32]. Several images of the target are captured from different viewpoints and orientations. Based on these images, the camera parameters can be estimated with relatively small calibration errors. For this reason, target-based calibration methods are commonly used in many computer vision applications.

2.3.1. Target-Based Camera Calibration

There are various approaches for target-based camera calibration. One of the most common methods uses a checkerboard pattern, as mentioned before. The checkerboard consists of a regular grid of black and white squares with known dimensions, as shown in Figure 4.

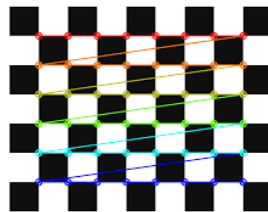


Figure 4: Example of a checkerboard and detected corner points. Source: [14].

During the calibration process, control points are extracted from the images. In the case of a checkerboard, these control points correspond to the corner points of the squares [25]. The positions of these corners are known in the coordinate system of the calibration target. By capturing multiple images of the checkerboard from different angles and distances, a set of 2D image points can be obtained that correspond to the known 3D points of the target. Based on these point correspondences, an optimization problem can be formulated to estimate the intrinsic and extrinsic camera parameters as well as the lens distortion coefficients [32].

This calibration approach is implemented in widely used computer vision libraries such as OpenCV [20]. It can be applied to single-camera setups and multi-camera systems. Besides checkerboards, other types of calibration targets exist, for example circular dot patterns [25]. In some cases, these patterns can achieve higher accuracy, especially when the image quality is low. However, they also require physical targets and a manual calibration setup.

The main limitations of target-based calibration methods are the required manual placement of the target and the need to capture images with different poses. In addition, the calibration accuracy depends on factors such as the quality of the corner detection, the number of images used, and the diversity of viewpoints during data acquisition [32]. Despite these limitations, target-based calibration remains a standard method in many computer vision applications due to its simplicity and reliable performance.

2.3.2. Calibration Parameters and Error Metrics

To evaluate the quality of a camera calibration, different error metrics are used. These metrics describe how well the estimated camera parameters fit the observed image data. In practice, this is done by comparing detected image points with points that are projected using the calibrated camera model. The most common metric is the reprojection error [19].

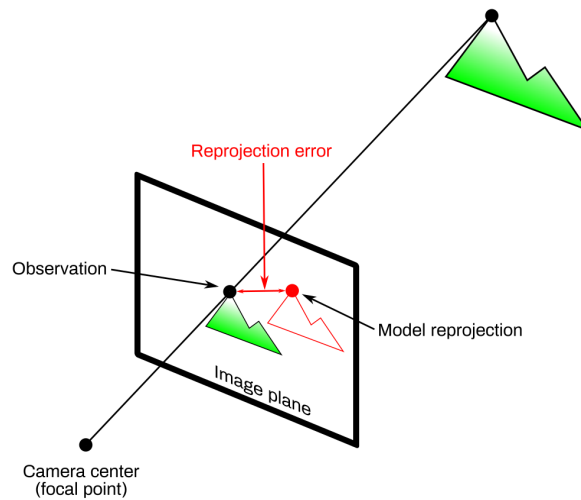


Figure 5: Visualization of reprojection error. Source: [2].

After a calibration, known 3D points are projected back onto the image plane. The reprojection error measures the distance between the observed image points and the

corresponding projected points. It is usually computed as the average Euclidean distance and is given in pixel units [2]. It can be defined as

$$e = \frac{1}{N} \sum_{i=0}^{N-1} \|\mathbf{p}_i - \mathbf{q}_i\|_2. \quad (7)$$

Here, $\mathbf{p}_i$ represents the observed feature points in the image, while $\mathbf{q}_i$ denotes the projected points obtained from the calibrated camera model. In general, a smaller reprojection error indicates a better calibration. What is considered a good value depends on the application, but for accurate calibrations the reprojection error is often between 0.1 and 0.5 pixels [12].

Another commonly used measure is the root mean square (RMS) error [11]. It summarizes the reprojection error over all calibration points and is computed as:

$$\epsilon_{\text{res}} = \left(\frac{1}{2n} \sum_{i=1}^n \|\mathbf{p}'_i - \mathbf{q}'_i\|_2^2 \right)^{1/2} \quad (8)$$

The RMS error provides a single value that describes the overall calibration accuracy and is often used to compare different calibration results.

2.3.3. Bundle Adjustment

Bundle adjustment is a method for improving a 3D reconstruction by refining both the positions of 3D points and the camera parameters at the same time [27]. The name comes from the “bundles” of light rays from the 3D points to the cameras, which are adjusted together to make the reconstruction more accurate. The parameters that are optimized together usually include the 3D coordinates of the points, the extrinsic parameters, and sometimes the intrinsic parameters [4].

This joint optimization improves calibration accuracy because it ensures that all cameras and points are consistent with each other. As a result, projection errors are reduced and small misalignments are corrected that would remain if cameras or points were adjusted separately [27]. This idea is illustrated in Figure 6.

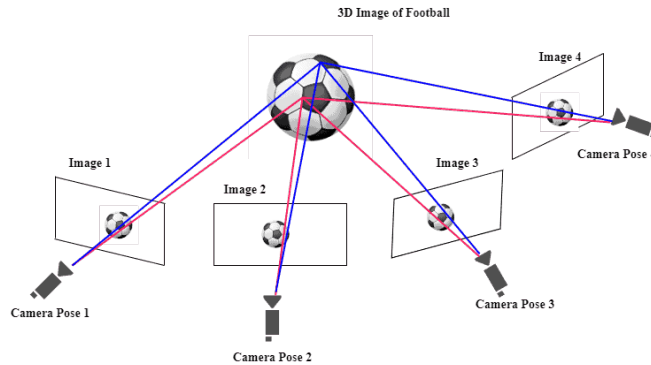


Figure 6: Visual representation of bundle adjustment. Source: [15].

Mathematically, bundle adjustment is formulated as an optimization problem that minimizes the reprojection error. This optimization is usually solved using nonlinear least squares methods and takes advantage of the fact that the problem structure is largely sparse [4].

2.4. Light Field Camera Systems

A standard camera captures a 2D projection of the 3D world. Light field cameras, on the other hand, have the ability to capture more information. They record not only the intensity of light rays but also their direction [10]. This property allows the creation of accurate depth maps and refocus the image after it has been taken [8]. Therefore light field cameras are growing in importance for applications in computer vision, robotics, augmented reality, and other applications.

The four-dimensional nature of light fields means that their processing currently needs relatively high computational resources [10]. However, the evolution of hardware and algorithms is making light field cameras more practical for real-world applications.

There are mainly two different types of light field camera systems: multi-camera arrays and micro-lens-array-based cameras. Each type has its own advantages and disadvantages, which will be discussed in the following sections.

2.4.1. Multi-Camera Light Field Systems

Multi-camera arrays are one way to capture light fields. This type of system is structured by arranging multiple individual cameras in a geometric configuration [29]. Each camera captures a different viewpoint of the scene, enabling the system to record a 4D light field. In practice, there are different approaches to arranging the cameras. For example, in [26], an array of 128 cameras in a square grid is used, while in other works cross-shaped structures with fewer cameras are employed [29], as shown in Figure 7.

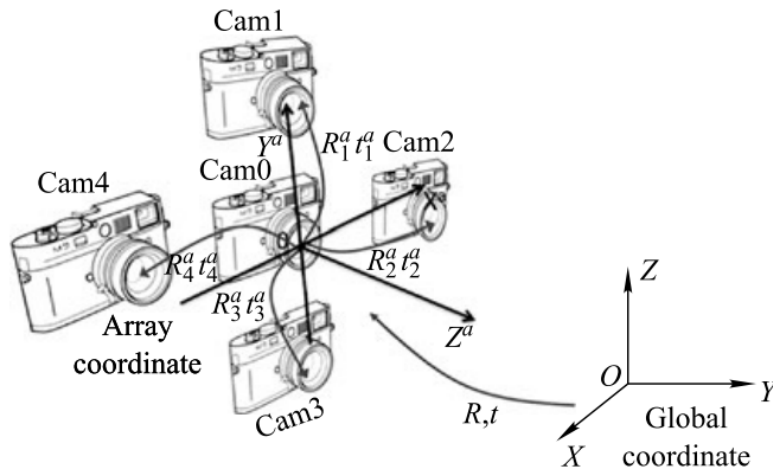


Figure 7: Representation of a multi-camera system. Source: [29].

Besides the advantages of a multi-camera setup, such as improved depth accuracy and refocusing capabilities, there are also challenges. The calibration of such systems is more

complex and requires higher computational effort due to the larger number of cameras involved [29].

It is also essential to keep the cameras properly aligned and synchronized. Even small errors in alignment or timing can cause artifacts or undesired results in the final image. Therefore, multi-camera light field systems require a very precise setup and calibration to work as intended.

2.4.2. Micro-Lens-Array-Based Light Field Cameras

The second type of light field camera is based on a micro-lens array (MLA). In this case instead of having different cameras, the system uses only one sensor combined with an array of small lenses placed between the main lens and the image sensor [8]. The MLA allows the camera to capture also the 4D light field information.

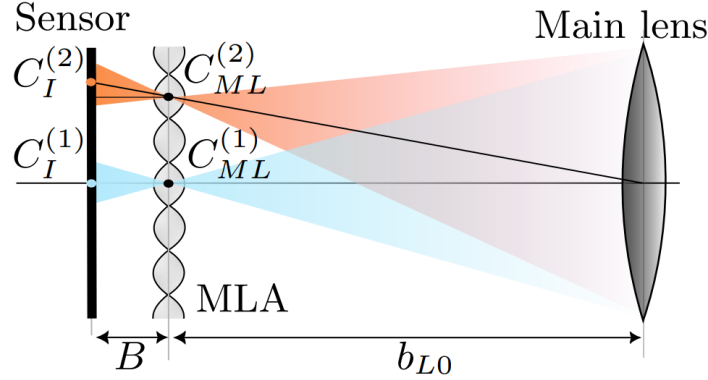


Figure 8: Structure of a micro-lens-array-based light field camera. Source: [8].

As illustrated in Figure 8, each lens in the MLA focuses light from different directions onto different regions of the sensor. This way, the camera is also able to capture the same information as a multi-camera array, but with a more compact design. Because of the smaller size and lower cost, MLA-based light field cameras are mostly used in consumer applications [16]. However, microlens-array cameras have to compromise between spatial and angular resolution, while multi-camera systems capture full-resolution images with sharper results and more accurate depth information.

For MLA-based light field cameras, the pattern-based calibration method provides satisfactory results in terms of reprojection error. In practice, the overall calibration quality strongly depends on the calibration target, particularly its flatness [1].

2.5. State of the Art in Light Field Camera Calibration

Accurate calibration is a central requirement for light field camera systems, as even small geometric errors directly affect depth estimation and view consistency. In camera arrays, where many individual views must agree, calibration errors accumulate and lead to visible artifacts in depth maps and reconstructed images.

Early work on light field calibration mainly relies on pattern-based approaches. Dima et al. systematically evaluate several multi-camera calibration methods using ground truth

data and show that calibration accuracy strongly depends on the array geometry and the physical setup of the calibration target [5]. In particular, small deviations such as bending or uneven surfaces of the calibration board already lead to increased reprojection errors across the array. Yang et al. demonstrate that using a simple checkerboard target can still yield stable calibration results for camera arrays while reducing setup complexity [30]. However, the accuracy of these methods remains closely tied to target quality and careful placement.

To overcome the limitations of target-based calibration, several self-calibration approaches have been proposed. Zhang et al. introduce a fixed relative pose prior to stabilize self-calibration in camera arrays [31]. By constraining relative camera poses, their method reduces pose drift in situations where image features are sparse or unevenly distributed. Zhou et al. address low-texture sports scenes and use temporal fusion across multiple frames to estimate calibration parameters [33]. These approaches reduce the need for calibration targets, but still depend on sufficient scene structure and a reasonable initial alignment.

Some methods focus on specific system layouts or downstream tasks. Ge et al. study a ring-shaped multi-camera system and use a transparent calibration target to support panoramic measurements [9]. Xie et al. show that accurate camera array calibration is essential for reliable depth map fusion, as calibration errors propagate directly into depth artifacts, especially near object boundaries [28].

The approach most relevant to this thesis is LiFCal by Fleith et al. [8]. Instead of using a physical calibration target, LiFCal moves the MLA-based camera through the scene and estimates calibration parameters via bundle adjustment driven by depth maps from multiple viewpoints. This shifts the calibration problem from controlled target acquisition to scene structure and camera motion, making the method particularly suitable for scenarios where calibration targets are impractical or unavailable.

Overall, existing work shows that calibration accuracy in light field systems remains sensitive to setup quality, motion, and scene structure. Larger camera arrays amplify small errors, and target-free methods require reliable depth estimates and sufficient camera motion to properly constrain the geometry.

3. Methods and Materials

This chapter introduces the methods and materials used in this thesis, covering both the design of the calibration framework and the technical setup used for its implementation and evaluation.

3.1. Methods

In this first section, the design and implementation of the calibration software are described. It describes the overall workflow, the use case scenario, and the specific requirements that guided the development process. The UML class diagram and state diagram illustrate the software architecture and behavior.

3.1.1. Overall approach

The goal of this thesis is to develop and evaluate an automatic data-driven calibration framework for multi-camera light field systems based on the target-free calibration method proposed by Fleith et al. [8]. As discussed in the state of the art, classical target-based calibration approaches require manual intervention and careful placement of calibration targets, which limits their applicability in long-term deployments or scenarios with restricted human access.

The proposed approach uses a reliable target-based calibration as an initial reference and ground truth. This initial calibration is performed using a checkerboard pattern and serves as a baseline for evaluating subsequent target-free recalibration steps. After this initial setup, the system captures a small set of images from the scene and applies LiFCal to automatically recalibrate the multi-camera system. To assess whether the target-free recalibration is acceptable, a health monitoring mechanism is introduced. This mechanism compares the calibration results obtained from LiFCal with the baseline calibration using defined decision metrics. If the recalibration achieves a calibration quality comparable to or better than the initial calibration, it is accepted and replaces the previous baseline. Otherwise, the system retains the existing calibration parameters.

This workflow is designed to enable periodic self-recalibration without manual intervention. A potential application scenario is a robotic system deployed in a remote or inaccessible environment, where mechanical shifts or environmental changes may degrade calibration quality over time. By regularly capturing image sets and applying the proposed recalibration process, the system can maintain accurate camera parameters throughout its operation.

To clearly describe the interaction between the individual components and calibration phases, a use case description is introduced in the following section.

3.1.2. Use Case Description

The following use case describes the intended workflow of the proposed automatic calibration framework for a cross-shaped multi-camera light field system consisting of a central camera and twelve surrounding cameras, arranged in four cardinal directions (North, East, South, and West). The workflow is structured into three consecutive phases: an

initial reference calibration, periodic target-free recalibration, and a health-based decision step.

Phase 1 — Initial Reference Calibration (once)

- **Target-based calibration (OpenCV):** A one-time checkerboard-based calibration is performed to estimate the intrinsic and extrinsic parameters for all cameras of the cross-shaped array.
- **Optional refinement:** An optional global refinement using bundle adjustment is applied to improve consistency across all camera parameters.
- **Baseline creation:** A global health score is computed for the initial calibration and stored as `baseline.json`. This baseline represents the known-good reference calibration for subsequent recalibration decisions.
- **Logging:** The complete initial calibration is stored as a timestamped JSON file and a summary entry is appended to `calibration.log`.

Phase 2 — Automated Recalibration Attempts (periodic scheduler)

- **Trigger:** An automated recalibration cycle is executed at fixed time intervals (e.g. every five minutes).
- **Conditional data preparation:** If LiFCal-compatible data (focus and depth directories) already exists and is complete, it is reused. Otherwise, the system generates the required inputs by running `run_imageset_creation.py`.
- **LiFCal recalibration run:** The script `run_LiFCal_calibration.py` executes LiFCal-based recalibration for all cameras in the array and produces per-camera parameter files as well as a combined multi-camera result.
- **Logging and traceability:** The combined recalibration result is stored as a timestamped JSON file and all relevant execution details are appended to `calibration.log`.

Phase 3 — Health Evaluation and Baseline Update Rule

- **Health score computation:** A single global health score in the range $[0, 100]$ is computed from the combined LiFCal calibration result.
- **Comparison against baseline:** The newly computed health score is compared to the score stored in the current baseline calibration.
- **Acceptance rule:** If the new health score is strictly higher than the baseline score, the baseline calibration is replaced by the new result. Otherwise, the existing baseline remains unchanged.
- **Continuous improvement loop:** The recalibration process is repeated periodically, allowing improved calibration results to update the baseline over time without manual intervention.

This structure allows the system to maintain accurate calibration parameters over time, adapting to changes in the environment or camera setup. The health-based decision mechanism ensures that only beneficial recalibrations are accepted, preserving the integrity of the calibration throughout the system's operation.

This use case serves as the conceptual basis for the requirements analysis and system design presented in the following sections.

3.1.3. Requirements Analysis

The requirements for the proposed automatic calibration framework are divided into functional and non-functional requirements. These requirements define the necessary features and qualities of the system to achieve the intended use case.

Functional Requirements

- **FR1 – Initial Reference Calibration (Target-based):**
The system shall perform a one-time, checkerboard-based initial calibration using OpenCV to estimate intrinsic and extrinsic parameters for the cross-shaped camera array. This initial calibration shall be stored as the baseline calibration and serve as the reference state for all subsequent health evaluations and recalibration decisions.
- **FR2 – Parameter Export and Standardized Representation:**
The system shall export all calibration results (initial calibration and LiFCal-based recalibration) in a standardized, machine-readable JSON format, including intrinsic parameters, extrinsic poses, reprojection error statistics, timestamps, and metadata.
- **FR3 – Periodic Calibration Health Monitoring:**
The system shall periodically (at least once per configured scheduler interval) evaluate the calibration quality by computing a global health score.
- **FR4 – Health Score Computation (Global Scalar Metric):**
The system shall compute a single scalar health score in the range $[0, 100]$ representing the overall calibration quality of the complete camera array. For the initial calibration, the health score shall be derived from reprojection error statistics. For LiFCal-based recalibration, the health score shall be derived from LiFCal-specific reprojection and consistency metrics extracted from the combined parameter set.
- **FR5 – Target-Free Online Recalibration (LiFCal-based):**
The system shall execute a LiFCal-based target-free recalibration using automatically prepared focus images and virtual depth data whenever a recalibration cycle is triggered.
- **FR6 – Acceptance and Baseline Update Rule:**
The system shall compare the health score of a newly computed calibration against the health score of the currently stored baseline calibration. The new calibration shall be accepted and stored as the new baseline only if its health score is strictly higher than the baseline score; otherwise, the system shall retain the previous baseline calibration.

- **FR7 – Automated Periodic Recalibration Scheduling:**
The system shall support automated, periodic execution of the LiFCal-based recalibration process at a configurable time interval (minimum every 5 minutes).
- **FR8 – Logging and Traceability:**
The system shall log every calibration and recalibration run, including at least timestamps, used image sets, computed health scores, baseline update decisions, and error states.
- **FR9 – Abort and Logging on Recalibration Failure:**
The system shall abort the current recalibration cycle if a recalibration run fails. In such cases, the system shall retain the currently active baseline calibration and record a descriptive error message, including the failure reason, in the calibration log.

Non-Functional Requirements

- **NFR1 – Modularity:**
The system shall be structured into modular components (initial calibration, LiFCal recalibration, health monitoring, scheduling, and logging) to support extensibility and maintainability.
- **NFR2 – Reproducibility:**
All calibration results and health evaluations shall be reproducible using stored JSON files and logged metadata.
- **NFR3 – Robustness:**
The calibration and monitoring pipeline shall tolerate moderate variations in scene content and illumination without system failure, even if recalibration quality temporarily degrades.
- **NFR4 – Automation:**
The calibration framework shall operate fully automatically after initial setup, without requiring manual intervention during periodic monitoring and recalibration.
- **NFR5 – Transparency:**
The system shall provide access to intermediate calibration metrics and logged decision outcomes to allow inspection and analysis of health score computations and acceptance decisions.

These requirements form the basis for the system design and implementation presented in the following sections. They were derived following the guidelines and best practices of IEEE Std 830–1998 [13] and directly support the intended use case of the proposed calibration framework. The functional requirements are reflected in the architectural decisions, UML diagrams, and implementation steps, while the non-functional requirements address relevant quality aspects such as robustness, automation, and maintainability.

The requirements specification is intentionally kept compact and focused. It is not intended to represent a complete industrial requirements document, but rather a research-oriented specification that is appropriate for the scope and objectives of this bachelor thesis.

Based on the requirements analysis and the defined use case, a UML class diagram was designed to structure and illustrate the main components of the project. The diagram in Figure 9 shows the key classes, their attributes, methods, and relationships.

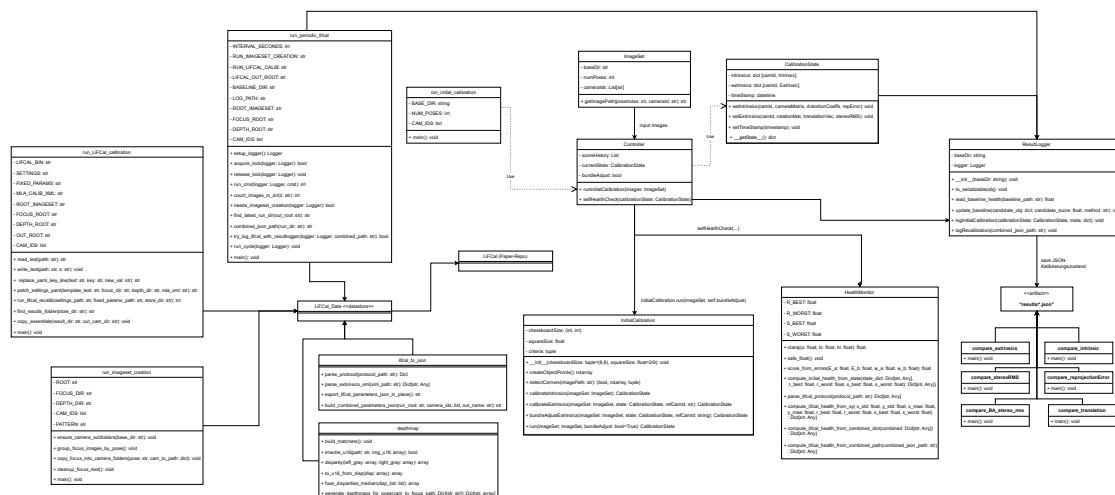


Figure 9: UML Class Diagram

The software architecture is based on a modular design, so that individual components can be developed, tested, and maintained independently. If future extensions or modifications are needed, the structure allows the easy integration of new features without major changes to the existing codebase.

The core class is the **controller**, which manages the overall calibration workflow. It coordinates the initial calibration, periodic recalibration attempts, health monitoring, and logging. The controller interacts with several classes that contain the specific functionalities.

The initial calibration and periodic recalibration are implemented in the `InitialCalibration` and `run_LiFCal_calibration` classes, respectively. To run these classes correctly, the `run_initial_calibration`, `run_periodic_lifcal`, and `run_imageset_creation` classes are responsible for preparing the required data and executing the calibration processes. In addition, the `depthmap` class creates, based on the image set, a depth map for each image, which is used for the LiFCal calibration. For health monitoring and decision-making, the `HealthMonitoring` class computes the health scores and applies the acceptance rules based on the defined criteria. These criteria will be discussed in the next section.

The `LiFCal` class implements the target-free calibration method proposed by Fleith et al. [8]. This class encapsulates the core logic of LiFCal, including the bundle adjustment optimization. It provides methods to run the calibration process and retrieve the estimated camera parameters.

Finally, for results and traceability, the `ResultLogger`, `lifcal_to_json`, and `CalibrationState` classes handle the logging of calibration runs, health evaluations, and decisions. This ensures that all relevant information is recorded in a structured format for later analysis.

3.1.5. Health Monitoring and Decision Metrics

As mentioned in the use case description, a key component of the proposed calibration framework is the health monitoring mechanism. This mechanism evaluates the quality of each recalibration attempt and decides whether to accept or reject the new calibration based on defined metrics. To define these metrics, the following formulas are used:

$$\text{HealthScore} = 100 \cdot \left(1 - \left(0.6 \cdot E_{\text{reproj}}^2 + 0.4 \cdot E_{\text{rms}}^2\right)\right) \quad (9)$$

$$E_{\text{reproj}} = \text{clamp}\left(\frac{e_{\text{reproj}} - R_{\text{BEST}}}{R_{\text{WORST}} - R_{\text{BEST}}}, 0, 1\right) \quad (10)$$

$$E_{\text{rms}} = \text{clamp}\left(\frac{e_{\text{rms}} - S_{\text{BEST}}}{S_{\text{WORST}} - S_{\text{BEST}}}, 0, 1\right) \quad (11)$$

This health score formulation is used for the initial target-based calibration. It combines normalized reprojection error and RMS error into a single scalar score in the range $[0,100]$, where higher values indicate better calibration quality. Clamp functions ensure that the error terms are bounded between 0 and 1 based on predefined best and worst-case thresholds. The weights are chosen to give slightly higher priority to the reprojection error than to the RMS error. This choice follows common practice in the literature, where reprojection error is treated as a primary indicator of calibration correctness and overall geometric accuracy.

For the LiFCal-based recalibration, a similar health score is computed using LiFCal-specific metrics:

$$\text{HealthScore}_{\text{LiFCal}} = 100 \cdot \left[1 - \left(0.6 \cdot E_{\text{std}}^2 + 0.4 \cdot E_{\text{mae}}^2\right)\right] \quad (12)$$

$$E_{\text{std}} = \text{clamp}\left(\frac{\sqrt{x_{\text{std}}^2 + y_{\text{std}}^2} - S_{\text{BEST}}}{S_{\text{WORST}} - S_{\text{BEST}}}, 0, 1\right) \quad (13)$$

$$E_{\text{mae}} = \text{clamp}\left(\frac{\sqrt{x_{\text{mae}}^2 + y_{\text{mae}}^2} - R_{\text{BEST}}}{R_{\text{WORST}} - R_{\text{BEST}}}, 0, 1\right) \quad (14)$$

with

$$R_{\text{BEST}} = 0.03, \quad R_{\text{WORST}} = 1.0, \quad S_{\text{BEST}} = 0.0, \quad S_{\text{WORST}} = 50.0$$

This formulation is applied to LiFCal-based recalibration results. It evaluates the stability and consistency of the estimated camera parameters using standard deviation and mean absolute error metrics. The resulting health score enables a direct comparison with the baseline calibration to support automated acceptance or rejection decisions. The initial and recalibration health scores are computed differently due to the different result metrics provided by the respective calibration methods. The specific best and worst-case values are defined also based on calibration performance which we discussed in Calibration Parameters and Error Metrics section.

3.1.6. State Diagram

In order to illustrate the workflow and interactions between the different calibration phases and components, a state diagram is presented in Figure 10. This diagram visualizes the transitions between the initial calibration, periodic recalibration attempts, and health evaluation steps.

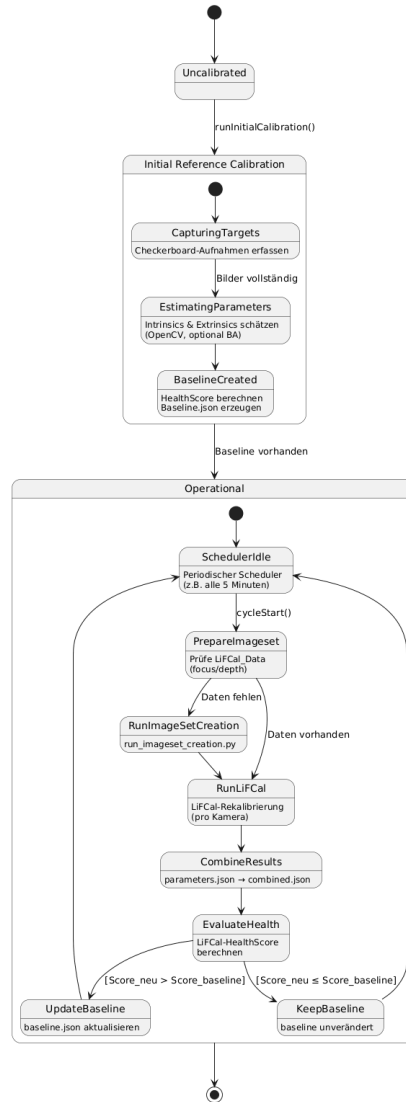


Figure 10: State Diagram

3.2. Materials

In this chapter, the materials and tools used for the implementation and testing of the calibration software are described and presented. Due to the lack of a physical multi-camera light field system, synthetic data generation with Blender was used to create realistic test scenarios. The structure of the image sets, ground truth definitions, and the runtime environment are also discussed.

3.2.1. Runtime Environment and Toolchain

Most parts of the implementation are based on established open-source libraries and tools. The primary programming language used is Python, chosen for its extensive ecosystem supporting computer vision, numerical computation, and scientific prototyping. Core libraries include OpenCV for calibration and image processing, NumPy for numerical operations, and Matplotlib for visualization. The LiFCal implementation provided by Fleith et al. [8] was integrated into the workflow to perform target-free recalibration.

The development and evaluation were conducted on a system running Ubuntu 20.04.6 LTS. A complete list of software dependencies and library versions is documented in the project repository README files to support reproducibility. It should be noted that the LiFCal implementation relies on GPU acceleration and requires a compatible NVIDIA GPU to execute the optimization efficiently. Consequently, LiFCal-based recalibration cannot be performed on systems without suitable GPU support.

3.2.2. Synthetic Data Generation with Blender

To evaluate and test the proposed system, a world scene was simulated using open-source software. This software is Blender [3], which also considers physical properties of light, materials, and camera optics to generate realistic images.

The first step was to create the multi-camera array geometry. As described in the use case, a cross-shaped arrangement with one central camera and twelve surrounding cameras was constructed. Each camera was configured with the same sensor size and a focal length of 50mm to make the image sets as realistic as possible. The cameras were placed at equal distances from their neighboring cameras, forming a cross shape. Each camera was positioned on a grid to simulate the main structure of the array, as shown in Figure 11.

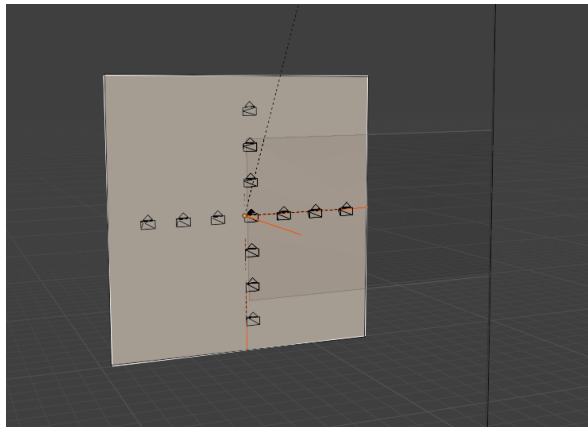


Figure 11: Multi-camera array setup in Blender.

The first scene that needed to be created included a calibration target for the initial calibration. Therefore, a checkerboard pattern was modeled and placed in front of a scene obtained from the Blender community [24]. The checkerboard was positioned in different locations and orientations to simulate a realistic calibration scenario. Due to the multi-camera setup, rendering the images took several minutes per pose, because each camera view had to be rendered individually. After rendering, the images were organized into folders according to the naming conventions required by the calibration scripts. The scene created for the initial calibration was modeled as a simple room, as shown in Figure 12.

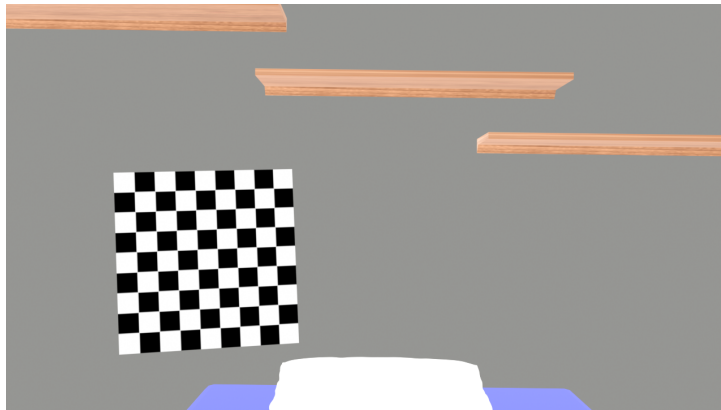


Figure 12: Example of a Blender scene for the initial calibration with a checkerboard.

For the LiFCal-based recalibration, a different scene was created without a calibration target. Since LiFCal relies on feature detection and depth maps, the scene had to be more complex and include a variety of objects and textures. Therefore, another scene from the Blender community was used [22], which represents a modern living room with furniture and decorations. The same camera setup was used, and the scene was rendered from different viewpoints to create the required image sets. An example of the scene used for LiFCal recalibration is shown in Figure 13.

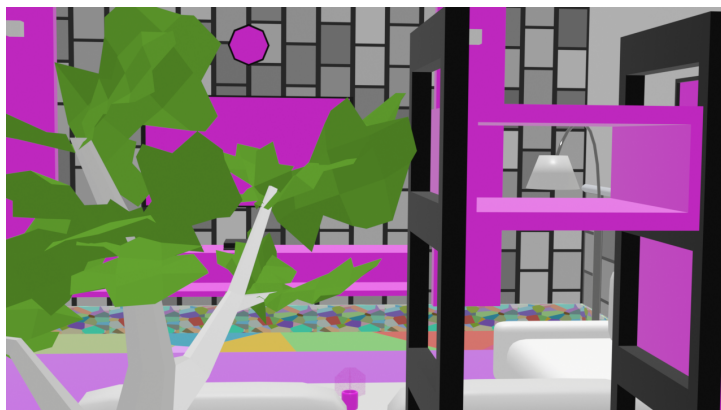


Figure 13: Example of a Blender scene for LiFCal recalibration without a calibration target.

The motivation behind using synthetic data was to maintain full control over the scene configuration and to simulate different calibration scenarios without the need for physical hardware. Another major advantage is the availability of ground truth camera parameters, which is essential for comparing the estimated calibration results with the actual values. This enables a more objective evaluation of calibration accuracy and the effectiveness of the implemented software.

3.2.3. Image Sets and Ground Truth Definition

To evaluate the calibration methods effectively, multiple image sets were created with different numbers of poses and camera views. In addition, for each image set, the multi-camera system was deliberately decalibrated to simulate real-world scenarios in which calibration quality may degrade over time due to mechanical or environmental influences.

The naming conventions for the image sets followed a structured and consistent format to ensure maintainability and usability. Each image set was organized into folders according to the calibration phase (initial calibration or LiFCal-based recalibration). File names encoded both the pose number and the camera identifier, for example `pose_001_Up1.png`.

The ground truth camera parameters were directly extracted from Blender. Since camera positions, orientations, and intrinsic parameters were explicitly defined within the 3D scene, the exact intrinsic and extrinsic parameters were known internally. The ground truth data was stored in JSON format, following the same structure as the calibration results, to enable direct and systematic comparison during evaluation. For each camera, both intrinsic parameters and extrinsic poses were included.

4. Results and Evaluation

In this chapter, the results, plots, and tables obtained from the implementation and from the comparison with the available ground truth data are presented. The evaluation focuses on the accuracy of the initial target-based calibration as well as on the performance of the LiFCal-based recalibration. The results are reported in terms of intrinsic and extrinsic parameter accuracy, reprojection errors, and the evolution of health scores.

4.1. Results of Initial Calibration

The results of the initial calibration are categorized into intrinsic accuracy, extrinsic accuracy, and reprojection and RMS error. Each category is discussed in the following subsections. To visualize the results effectively, the plots are taken from one specific image set in which the camera system was deliberately decalibrated by a certain degree to simulate real-world conditions. Additional plots for all recalibration runs are provided in subsection A.1.

4.1.1. Intrinsic Accuracy

The accuracy of the intrinsic parameters is evaluated by comparing the estimated focal lengths and principal points with the ground truth values obtained from Blender. The deviations in f_x , f_y , c_x , and c_y are shown in Figure 14.

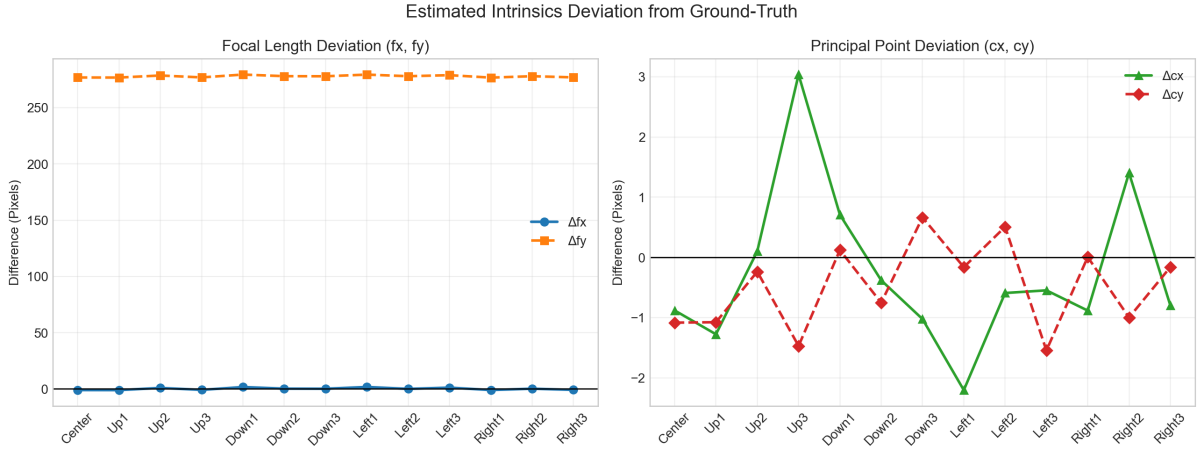


Figure 14: Deviations of intrinsic parameters from ground truth for the initial calibration.

The left plot shows the deviations of f_x and f_y for each camera in pixels. The blue curve represents the f_x deviation and the orange curve represents the f_y deviation. The estimated f_x values stay close to zero across all cameras, while the f_y values show a nearly constant offset of about 275 pixels.

The right plot presents the deviations of c_x and c_y , also in pixels. Here, the blue curve corresponds to c_x and the orange curve to c_y . Both parameters have small offsets from the ground truth, with most values remaining close to zero.

4.1.2. Extrinsic Accuracy

For the evaluation of the extrinsic parameters, the estimated camera poses are compared with the corresponding ground truth poses. In this case, rotation and translation errors are evaluated using two separate plots to clearly visualize the results, as shown in Figure 15.

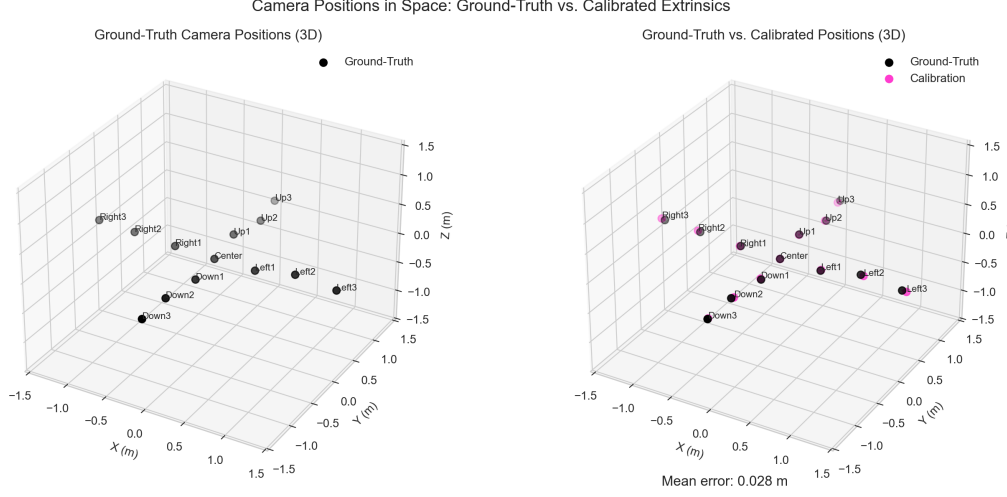


Figure 15: Translation parameter deviations from ground truth and the initial calibration.

In the left plot, the 3D positions of the multi-camera array from the ground truth are shown, while the right plot illustrates the estimated camera positions after the initial calibration. The mean translation error is shown in the lower part of the figure. The overall structure size is larger than that of a typical multi-camera array due to the synthetic setup in Blender. In both plots the main structure of the cross-shaped camera array is clearly visible.

The translation error per camera and the rotation error per camera are further evaluated in the plot shown in Figure 16.

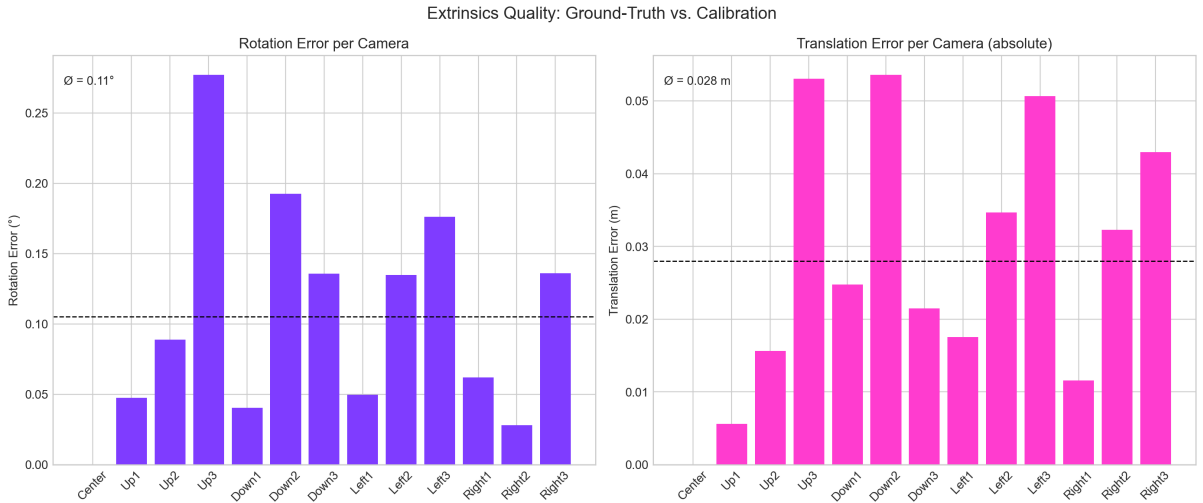


Figure 16: Extrinsic parameter errors of the initial calibration.

The left plot shows the rotation error in degrees for each camera, while the right plot shows the translation error in meters for each camera. The central camera was used as the reference camera in the calibration process, therefore, its errors are zero.

4.1.3. Reprojection and RMS Error

As discussed in the Calibration Parameters and Error Metrics section, reprojection error and RMS error are important indicators of calibration quality. The plot shown in Figure 17 presents the reprojection error per camera of the initial calibration.

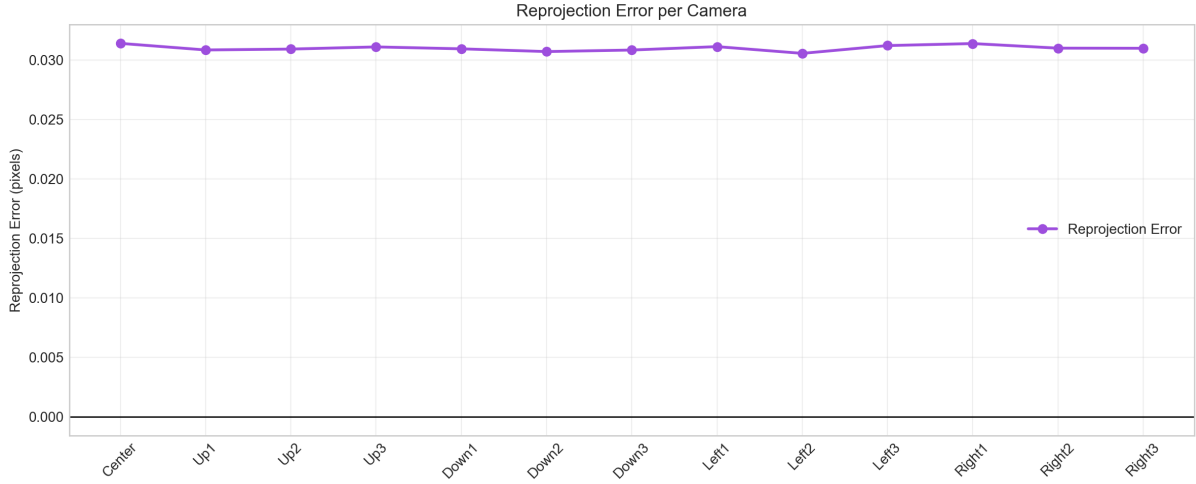


Figure 17: Reprojection error per camera of the initial calibration.

The RMS error for the multi-camera array is shown in Figure 18. These values are automatically provided by the OpenCV stereo calibration function.

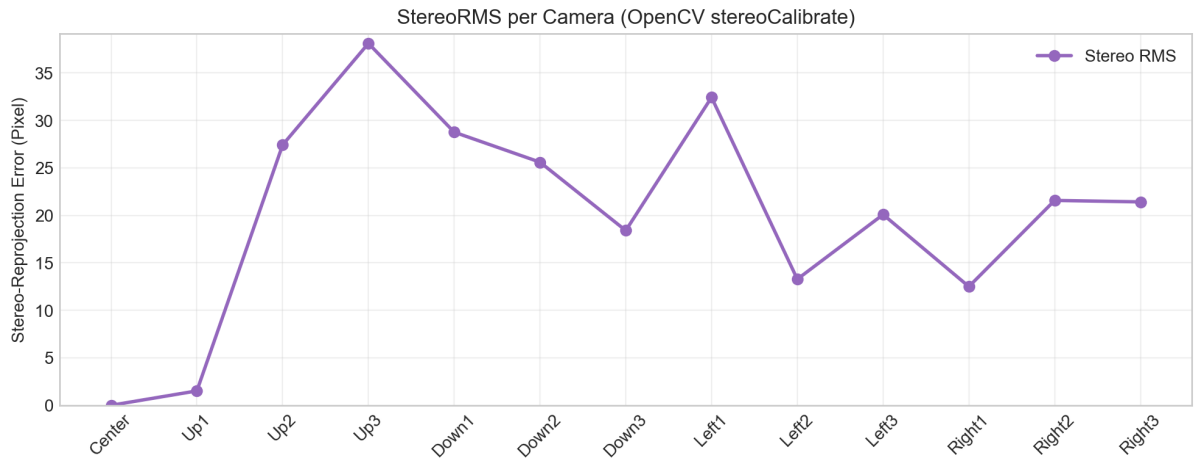


Figure 18: Stereo RMS error of the initial calibration.

Both the reprojection error and the RMS error are measured in pixels. The two plots were generated using the calibration results of the initial target-based calibration. In Figure 17 the error values are almost constant across all cameras with a reprojection error of about 0.3 pixels. In Figure 18 the RMS error values vary between 35 and 15 pixels for the different cameras.

4.1.4. Comparison of Mean Results Across Image Sets

To provide a consolidated overview of the initial calibration results, the mean values across different image sets are summarized in Table 1. The table reports the number of poses used for each image set, whether bundle adjustment (BA) was applied, as well as the mean translation error, mean rotation error, and mean reprojection error.

Name	#Poses	BA	Mean ΔT [m]	Mean ΔR [°]	Mean Reproj. [px]
imageset1	41	yes	0.07576	0.17447	0.33859
imageset2	33	yes	0.04710	0.16089	0.03274
imageset3	41	yes	0.09500	0.18225	0.40989
imageset4	17	yes	0.05414	0.23255	0.03103
imageset5	28	yes	0.02798	0.10531	0.03100
imageset6	36	yes	0.06228	0.20084	0.03070
imageset7	20	yes	0.04716	0.21424	0.03166

Table 1: Mean calibration results for different image sets

To analyze the calibration performance under different conditions, the image sets were generated with varying numbers of poses. The first four image sets (imageset1 to imageset4) are grouped into Scenario A_Robustness. In this scenario, the same multi-camera decalibration was applied in order to evaluate how robust the calibration results are with respect to different checkerboard poses.

The remaining image sets (imageset5 to imageset7) belong to Scenario B_Generalisation. In this case, different multi-camera decalibrations were applied to assess how well the calibration generalizes to different camera configurations with varying numbers of poses. A summary of both scenarios is provided in Table 2.

Scenario	#Runs	Avg. #Poses	Mean ΔT [m]	Mean ΔR [°]	Mean Reproj. [px]
A_Robustness	4	33.0	0.06800	0.18754	0.20306
B_Generalisation	3	28.0	0.04581	0.17346	0.03112

Table 2: Scenario-based summary of calibration results

4.2. Results of LiFCal-Based Recalibration

In this section, the results of the LiFCal-based recalibration are presented and visualized using plots. Due to the fundamentally different nature of the LiFCal calibration results compared to the available ground truth data, the evaluation focuses exclusively on reprojection error and the resulting health score.

A direct comparison of extrinsic parameters is not meaningful in this case. LiFCal requires multiple poses of the scene, with each pose resulting in a separate parameter estimation. Consequently, extrinsic parameters vary across poses and cannot be compared directly to a single ground truth configuration. In addition, LiFCal calibrates each camera independently, leading to camera parameters defined in individual local coordinate systems

rather than in a shared global reference frame. For these reasons, intrinsic and extrinsic parameters across cameras are not directly comparable in the same way as in the target-based calibration. These limitations are discussed in more detail in the Discussion of the Results section.

4.2.1. Reprojection Error after LiFCal

For the evaluation of the LiFCal-based recalibration, three independent recalibration runs using different image sets are considered. As a representative example, the reprojection error per camera for one selected run is shown in Figure 19. This visualization follows the same structure as the reprojection error plot for the initial target-based calibration shown in Figure 17. However, due to the output format of LiFCal, the reprojection error is reported separately for the x and y image directions.

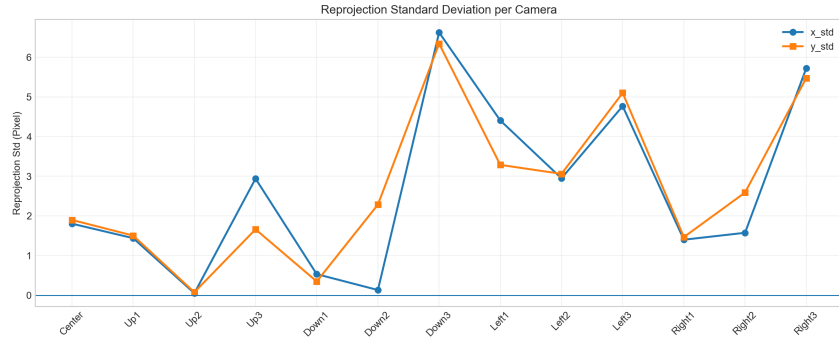


Figure 19: Reprojection error per camera for a single LiFCal-based recalibration run.

To assess the consistency of the reprojection error across multiple recalibration runs, Figure 20 shows the reprojection errors for all three runs. The upper part of the figure visualizes the x reprojection errors for each camera, while the lower part shows the corresponding y reprojection errors. This representation allows for a direct comparison of the reprojection error distributions across different runs and camera indices.

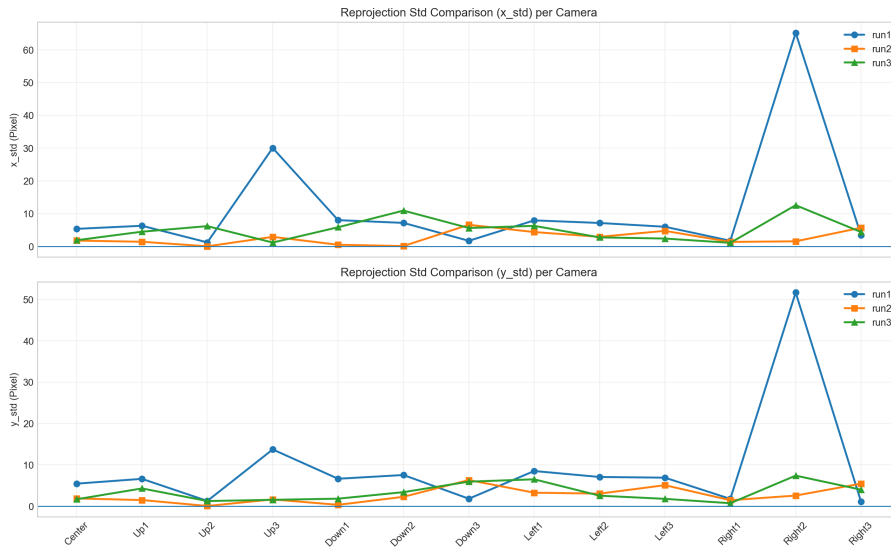


Figure 20: Reprojection error per camera across three LiFCal-based recalibration runs.

In addition, a direct comparison between the LiFCal-based recalibration and the initial target-based calibration is provided in Figure 21. The figure compares the mean reprojection errors in x and y for the best-performing LiFCal run with the reprojection error obtained from the initial calibration.

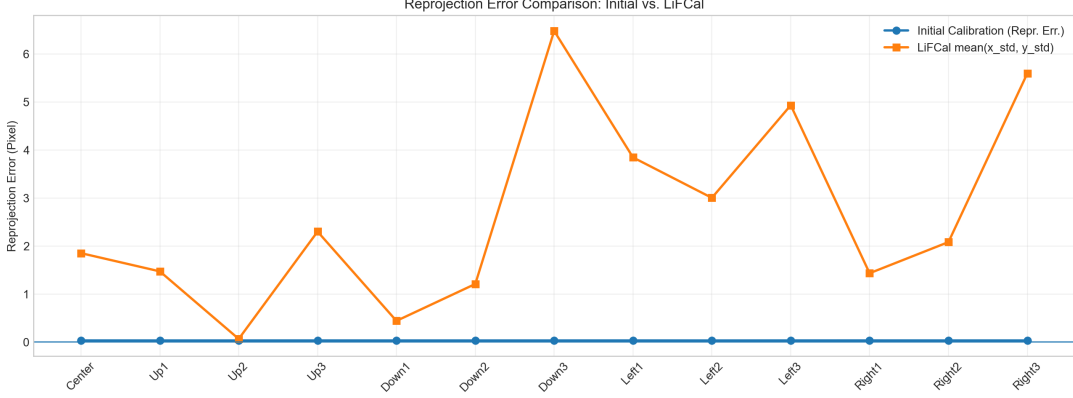


Figure 21: Comparison of mean reprojection error per camera between initial calibration and LiFCal-based recalibration.

The reprojection error of the initial target-based calibration remains close to zero for all cameras, with all values around 0.03px. After LiFCal-based recalibration, the reprojection error is higher for every camera. The values show a clear separation between the two calibration stages.

4.2.2. Health Score Comparison

The health scores are calculated using the formulas presented in the Health Monitoring and Decision Metrics section. All health scores for both the LiFCal-based recalibration runs and the initial target-based calibration image sets are summarized in Table 3. Note that the score range is $[0, 100]$, with higher values indicating better calibration quality.

Name	Calibration Type	#Poses	BA	Health Score
Run 1	LiFCal	4	true	52.0538
Run 2	LiFCal	10	true	59.4736
Run 3	LiFCal	20	true	58.7394
Imageset1	Initial	41	true	91.6138
Imageset2	Initial	33	true	97.8905
Imageset3	Initial	41	true	87.0450
Imageset4	Initial	17	true	97.1769
Imageset5	Initial	28	true	92.9879
Imageset6	Initial	36	true	88.6221
Imageset7	Initial	17	true	97.1303

Table 3: Overview of calibration runs and corresponding health scores

The first block of the table summarizes the health scores obtained from the three LiFCal-based recalibration runs, while the second block presents the health scores from the initial target-based calibration image sets. The health scores for the LiFCal runs are significantly

lower than those of the initial calibration. The highest health score among the LiFCal runs is approximately 59.47, while the highest health score from the initial calibration is around 97.89. The lowest health score from the initial calibration is 87.05, and the lowest from the LiFCal runs is 52.05.

5. Discussion of the Results

In this part of the thesis, the obtained results are analyzed and discussed in detail. The focus is on interpreting the findings from both the initial target-based calibration and the LiFCal-based recalibration. The limitations of the LiFCal method in the given setup are examined, and a comparison between the two calibration approaches is provided. The research questions outlined in the Introduction are also answered based on the results. Finally, potential extensions and improvements for future work are suggested.

5.1. Interpretation of Initial Calibration Results

The initial target-based calibration method provided very accurate results in terms of intrinsic and extrinsic parameter estimation. As shown in Figure 17, the reprojection error remained consistent across all cameras with a value of about 0.03 pixels. This indicates that the calibration process was able to accurately model the camera geometry and lens distortions. The intrinsic parameters, particularly the focal lengths and principal points, also showed small deviations from the ground truth values, as illustrated in Figure 14. In this case, the f_x values were very close to the ground truth, while the f_y values exhibited a nearly constant offset of approximately 275 pixels. This systematic deviation is due to the synthetic nature of the data and the specific configuration of the Blender camera model, because the sensor width and height were automatically set. Either the width or the height could have been adjusted and the other parameter would have been calculated accordingly to maintain the aspect ratio. This led to a consistent offset in one of the focal length parameters.

The RMS error values, shown in Figure 18, were not as low as expected, ranging from 15 to 35 pixels across the cameras. However, these values are not directly comparable to the per-camera reprojection errors, as the RMS returned by OpenCV’s `stereoCalibrate` represents a joint error over both camera views and also reflects the consistency of the estimated relative camera poses. Small inaccuracies in corner detection or checkerboard poses can therefore result in higher RMS values, especially before applying bundle adjustment.

The extrinsic parameters also demonstrated good accuracy, with mean translation errors around 0.068 meters and mean rotation errors approximately 0.187 degrees across different image sets, as summarized in Table 2. Almost all image sets achieved reprojection errors below 0.04 pixels, indicating a high level of geometric consistency in the estimated camera poses. However, in `imageset1` and `imageset3`, the reprojection errors were slightly higher, around 0.34 pixels and 0.41 pixels, respectively. These two anomalies are caused by the quality of the taken images during the rendering process in Blender. In these two image sets, the checkerboard distance to the camera was larger than in the other image sets, which made it more difficult to detect the corners accurately. This led to less precise calibration results and consequently higher reprojection errors. More poses with

closer checkerboard positions would likely have improved the calibration accuracy in these cases. That shows the importance of the variety and quality of the calibration images for achieving optimal results. A higher number of poses does not necessarily guarantee better calibration if the image quality is compromised.

In Table 2 it can be observed that the robustness and generalization capabilities of the initial calibration method are stable across different decalibration scenarios. Scenario A_Robustness, which tested the same decalibration with varying checkerboard poses, yielded consistent results with low mean translation and rotation errors. Scenario B_Generalisation, which involved different decalibrations, also produced reliable calibration outcomes, demonstrating the method’s ability to adapt to various camera configurations. This indicates that the initial target-based calibration is effective in maintaining accuracy under different conditions, provided that the calibration images are of sufficient quality.

5.2. Limitations of LiFCal in the Given Setup

The LiFCal-based recalibration results revealed several limitations when applied to the synthetic multi-camera light field setup. The visualization in Figure 21 clearly shows that the reprojection errors after LiFCal-based recalibration are significantly higher than those obtained from the initial target-based calibration.

However, it is important to note that LiFCal is designed as a target-free calibration method that relies on feature detection and depth maps. The synthetic data generated in Blender may not fully capture the complexity and variability of real-world scenes, which can negatively affect the performance of LiFCal. In addition, the depth maps used in the LiFCal process are derived from the synthetic scenes and may not accurately represent the depth information that would be obtained from real images.

A major reason for the high reprojection errors is that LiFCal was primarily developed for micro-lens-array (MLA) light field cameras rather than for multi-camera array systems. The assumptions made by LiFCal regarding the camera model and the image formation process do not fully hold for the multi-camera array setup used in this thesis. This mismatch can lead to inaccuracies in the estimated camera parameters and consequently to higher reprojection errors. Nevertheless, LiFCal provides a systematic target-free recalibration approach, which is a significant advantage in scenarios where the use of physical calibration targets is impractical.

Furthermore, as discussed in the use case description, the main purpose of the automatic recalibration software is to maintain calibration quality over time without manual intervention. Although LiFCal does not require a calibration target, it still depends on multiple images of the scene captured from different viewpoints. This implies that the camera system must be able to move within the scene in order to generate the required input data for LiFCal. In practical applications, this requirement may not always be feasible, particularly in static environments where the camera system cannot be repositioned easily. Therefore, while LiFCal enables a target-free calibration process, it still relies on a certain degree of controlled motion or data acquisition, which can limit its applicability in fully automated and static scenarios.

With regard to the first research question (RQ1) formulated in the Introduction—“*Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?*”—the results presented in this chapter indicate that LiFCal can be applied in a target-free manner, but its effectiveness in the given multi-camera light field setup is limited. The observed high reprojection errors suggest that LiFCal, in its current form, is not sufficient to achieve accurate calibration for this specific system without further adaptations. Nevertheless, the method provides a valuable foundation for an automated recalibration pipeline that can be improved and extended in future work.

5.3. Comparison between Initial Calibration and LiFCal

The comparison between the initial target-based calibration and the LiFCal-based recalibration highlights several key differences in terms of accuracy and limitations. As shown in Figure 20, the reprojection errors after LiFCal recalibration are inconsistent across the different runs, while the initial calibration produced stable and low reprojection errors across all cameras and image sets, as illustrated in Table 1. For the multi-camera array system used in this thesis and the synthetic data generated in Blender, the initial calibration method clearly outperformed LiFCal in terms of accuracy. The health scores summarized in Table 3 further emphasize this discrepancy, with the initial calibration achieving scores above 87, while the highest score obtained with LiFCal was only around 59. This substantial difference indicates that the initial calibration method is more effective at accurately modeling the camera system under the given conditions.

As mentioned earlier, the extrinsic parameters estimated by LiFCal cannot be directly compared to the ground truth data due to the independent calibration of each camera and the absence of a shared global reference frame. This limitation restricts the evaluation of the overall geometric consistency of the LiFCal results in the same way as for the initial calibration. Consequently, the comparison focuses primarily on reprojection errors and health scores, which provide a more indirect assessment of calibration quality.

With respect to the second research question (RQ2) posed in the Introduction—“*How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?*”—the findings show that LiFCal, in the current setup and within the design of the proposed software, does not perform well enough to replace a classical checkerboard-based calibration. The significantly higher reprojection errors and lower health scores indicate that LiFCal is currently unable to achieve the same level of accuracy as the initial calibration method. In addition, the observed outliers and inconsistencies in the LiFCal results suggest that further refinements and adaptations are necessary to improve its performance for multi-camera light field systems. Therefore, while LiFCal represents a promising approach, it cannot currently serve as a complete replacement for traditional calibration methods in this context.

5.4. Comparison to the State of the Art

The state of the art in multi-camera light field calibration is still dominated by target-based approaches that use physical calibration patterns (e.g., checkerboards) to estimate

intrinsic and extrinsic parameters reliably. The initial calibration implemented in this thesis follows this classical line of work and achieved consistently low reprojection errors (around 0.03px across cameras) as well as high health scores (roughly in the range of 87 to 98 across the evaluated image sets). Overall, these results are in line with what is typically reported for well-controlled target-based calibration in multi-camera systems.

Target-free approaches such as LiFCal are attractive because they avoid physical targets and, in principle, allow calibration updates with less manual setup. In the setup used in this thesis, LiFCal did not reach the same level of accuracy, as the reprojection errors were clearly higher across cameras, and the resulting health scores stayed far below those of the initial calibration (around 52 to 59 in the three runs). This suggests that, at least in the current form and for the considered multi-camera array scenario, LiFCal cannot replace the checkerboard-based baseline.

A practical contribution of this thesis to the state of the art is the automated calibration pipeline. It combines an initial reference calibration with periodic target-free recalibration attempts and a decision mechanism based on a health score. This directly addresses the real-world problem of calibration drift over time, where repeated manual target-based calibration can become impractical. Using Blender-based synthetic scenes also made it possible to run controlled experiments with access to ground truth parameters, which are difficult to obtain in physical multi-camera setups. This innovative approach provides a useful method for evaluating and comparing calibration techniques in a reproducible manner.

In summary, the initial target-based calibration matches established practice and produces strong results in this environment, while the LiFCal-based recalibration mainly shows potential but requires further adaptation before it can be considered a robust alternative for multi-camera light field systems.

5.5. Potential Extensions and Improvements

The current state of the implemented calibration software provides a solid foundation. However, there are several potential extensions and improvements that could be explored in future work to fulfill the goal of fully automated and reliable multi-camera light field calibration. One promising direction is to adapt and enhance the LiFCal method specifically for multi-camera array systems. This could involve modifying the underlying camera model assumptions and optimization strategies to better suit the geometry and characteristics of multi-camera setups.

An optional improvement would be to implement the software in a more performance-oriented programming language, such as C++ or Rust. This would allow faster execution of the calibration pipeline, which is particularly important for real-time applications or scenarios where frequent recalibration is necessary. Additionally, integrating parallel processing techniques could further speed up the calibration process, especially when dealing with large image sets or high-resolution images.

The evaluation of the calibration framework using a real multi-camera light field system would be a crucial next step. This would provide a more realistic assessment of the cali-

bration methods and their performance in practical scenarios. Testing the LiFCal method on real-world data would also help to identify additional challenges or limitations that may not be apparent in synthetic environments.

In order to explicitly answer the last research question (RQ3) from the Introduction—“*Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?*”—the following conditions and requirements can be identified based on the findings of this thesis:

- **LiFCal Adaptation:** The LiFCal method needs to be specifically adapted for multi-camera array systems to account for their unique geometry and characteristics.
- **Scene Complexity:** The scenes used for target-free recalibration should contain sufficient texture and depth variation to enable robust feature detection and depth estimation.
- **Camera Movement:** The camera system must be capable of capturing multiple viewpoints of the scene to provide the necessary data for LiFCal.
- **Quality of Input Data:** High-quality images with minimal noise and distortion are essential for accurate calibration results.
- **Computational Efficiency:** The calibration software should be optimized for performance to allow timely recalibration in practical applications.
- **Evaluation with Real Data:** Testing and validating the calibration methods using real multi-camera light field systems is crucial to ensure their effectiveness in practical scenarios.
- **Health Monitoring:** Implementing a robust health monitoring mechanism to assess calibration quality and decide when recalibration is necessary.

In conclusion, the requirements outlined in the Methods and Materials section and the potential improvements discussed here provide a guideline for future research and development in the field of data-driven automatic multi-camera light field calibration. Addressing these aspects will be key to achieving reliable and efficient target-free recalibration in practical applications.

6. Conclusion and Outlook

The final chapter summarizes the key aspects of the thesis. On one hand, the main achievements and findings are recapped in the Conclusion subsection. On the other hand, in the Outlook subsection, potential future directions and improvements for the calibration software and methods are discussed.

6.1. Conclusion

In this thesis, a data-driven automatic calibration software for multi-camera light field systems was implemented and evaluated. It combines an initial target-based calibration using a checkerboard, which is used to estimate the calibration parameters as a reference, with a target-free recalibration approach based on the LiFCal method. In addition, a health score monitoring mechanism is used to decide whether a new calibration result improves the overall calibration quality. The health score ranges from 0 to 100, where higher values correspond to better calibration quality. The system is designed to perform recalibration automatically, without requiring manual intervention, in order to counteract calibration drift over time. The software was implemented in Python, and synthetic image data generated with Blender was used to evaluate the calibration results against known ground truth parameters.

The results showed that the initial target-based calibration achieved high accuracy, with reprojection errors of approximately 0.03px per camera and health scores ranging between about 87 and 98 across the evaluated image sets. In contrast, the LiFCal-based recalibration resulted in significantly higher reprojection errors for all cameras and lower health scores, which ranged between approximately 52 and 59 in the performed recalibration runs. These results indicate that, in the given multi-camera light field setup, LiFCal is currently not suitable to replace a classical checkerboard-based calibration. The main limitations of LiFCal in this context were analyzed, and possible reasons for its reduced performance were discussed.

In conclusion, the implemented software provides a useful framework for automatic multi-camera light field calibration and systematic evaluation of calibration quality. However, further adaptations and improvements are required before the target-free recalibration approach can achieve a calibration accuracy comparable to that of the initial target-based method.

6.2. Outlook

For future work, several extensions of the calibration software were identified. One possible direction is to adapt the LiFCal method more explicitly to multi-camera array systems. This would require changes to the underlying camera model and the optimization process so they match the geometry and constraints of camera arrays more closely. Such adaptations may lead to more stable results when the method is applied to multi-camera configurations.

Another potential improvement concerns the implementation itself. Rewriting the software in a faster programming language, such as C++, would reduce runtime and make re-

peated recalibration more practical. This becomes relevant in real-world scenarios where recalibration is performed frequently or where time constraints play a role.

A further step would be the evaluation of the framework on a real multi-camera light field system. Testing on real hardware would allow a more realistic assessment of calibration accuracy and robustness. It would also expose practical issues which do not appear in synthetic experiments, such as noise, mechanical inaccuracies, or synchronization effects. Overall, the project presented in this thesis provides a solid foundation for further research and development in the field of multi-camera light field calibration.

A. Appendix

A.1. Additional Results

A.1.1. Imageset1

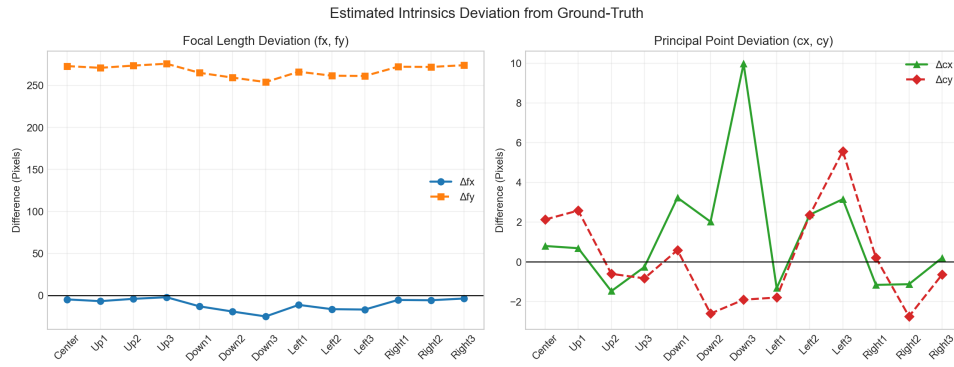


Figure 22: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset1).

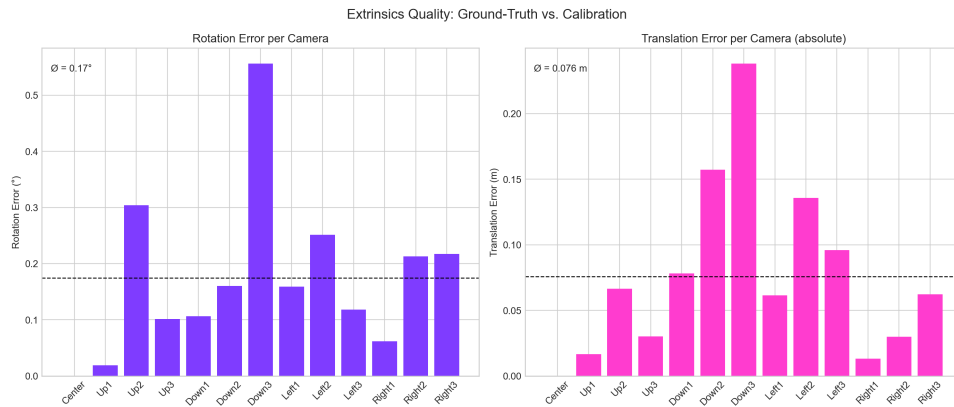


Figure 23: Extrinsic parameter errors of the initial calibration (Imageset1).

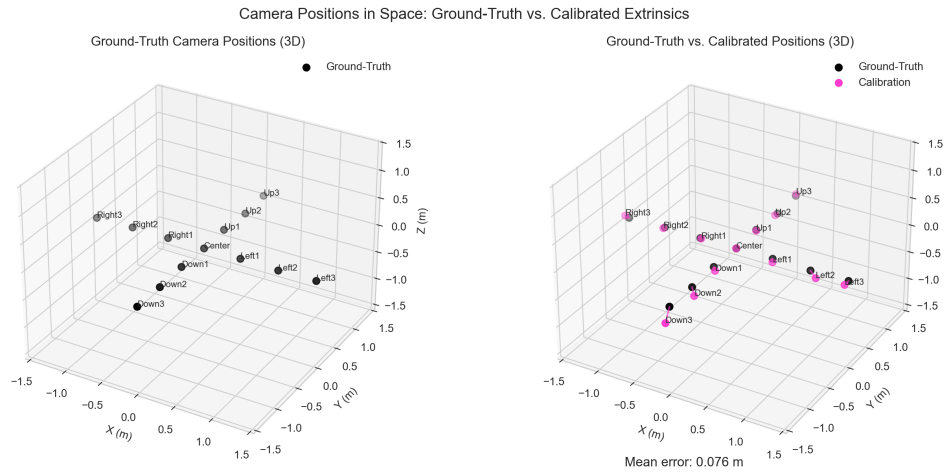


Figure 24: Translation parameter deviations from ground truth and the initial calibration (Imageset1).

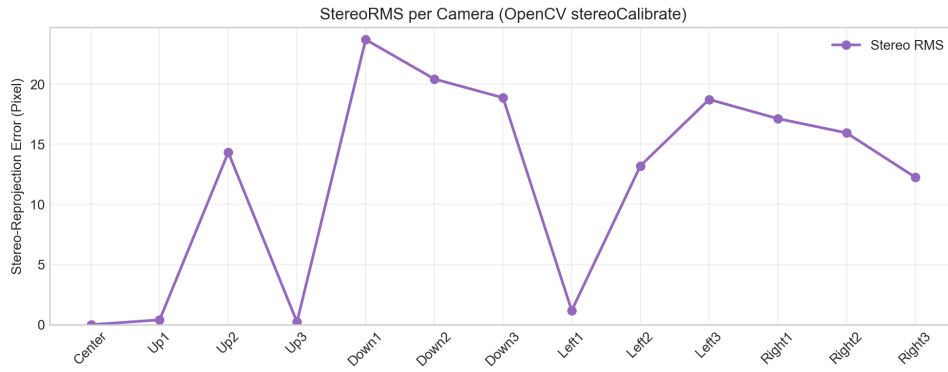


Figure 25: Stereo RMS error of the initial calibration (Imageset1).

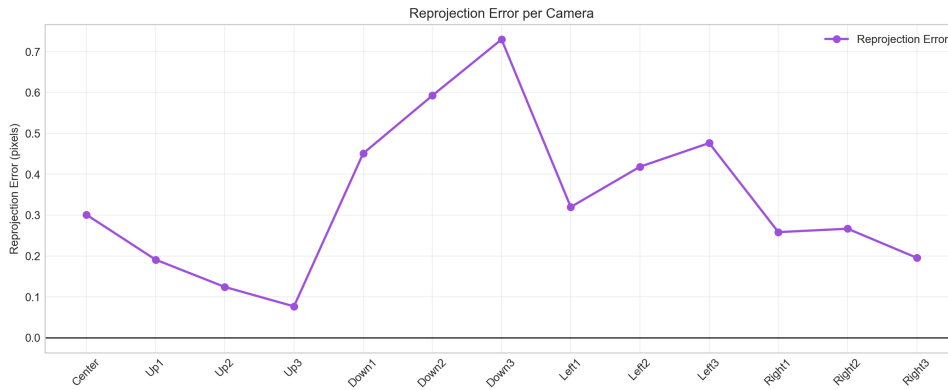


Figure 26: Reprojection error per camera of the initial calibration (Imageset1).

A.1.2. Imageset2

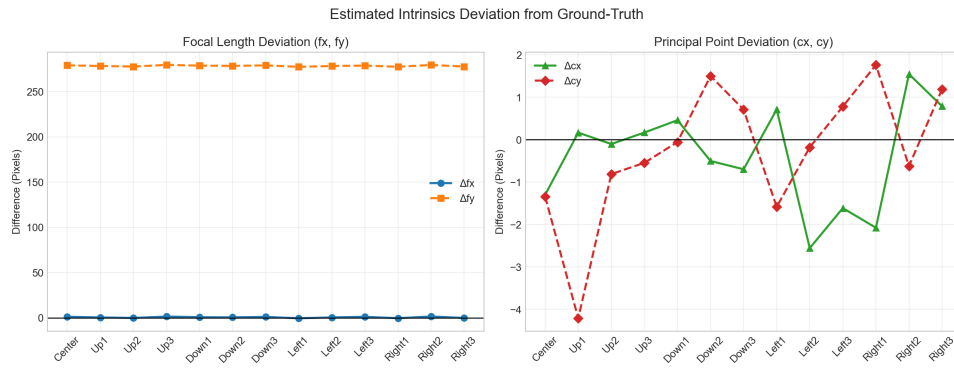


Figure 27: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset2).

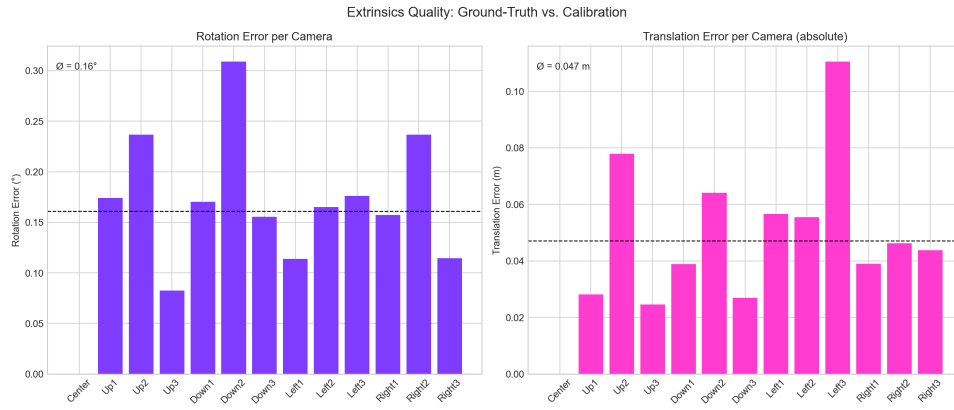


Figure 28: Extrinsic parameter errors of the initial calibration (Imageset2).

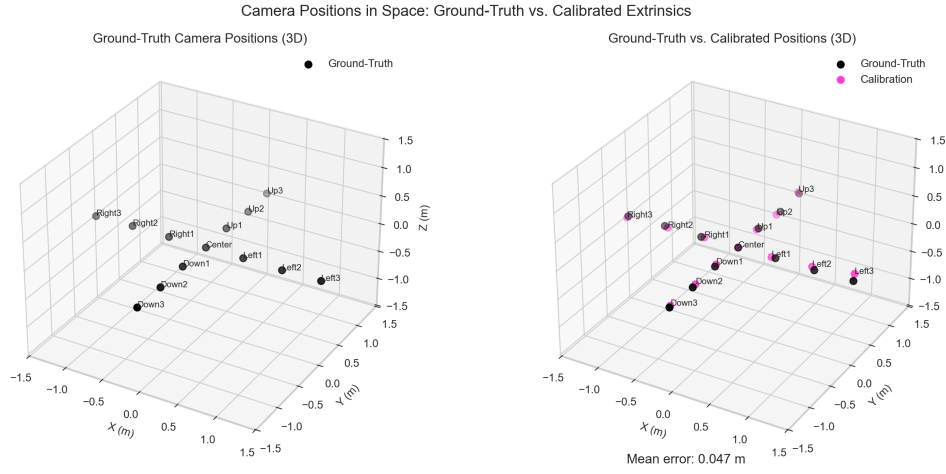


Figure 29: Translation parameter deviations from ground truth and the initial calibration (Imageset2).

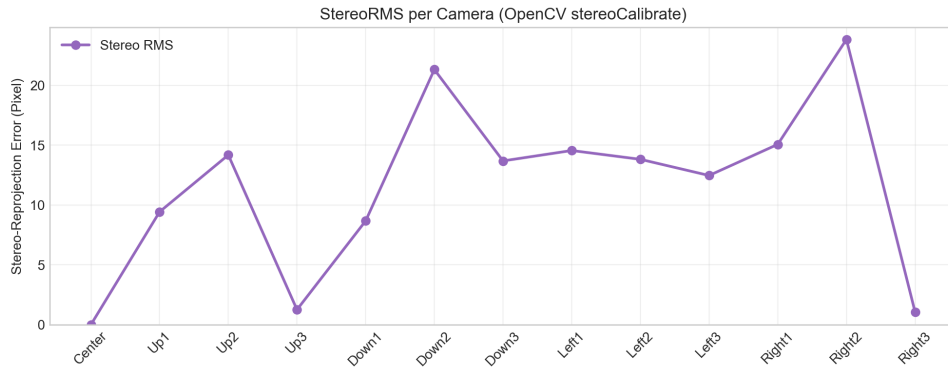


Figure 30: Stereo RMS error of the initial calibration (Imageset2).

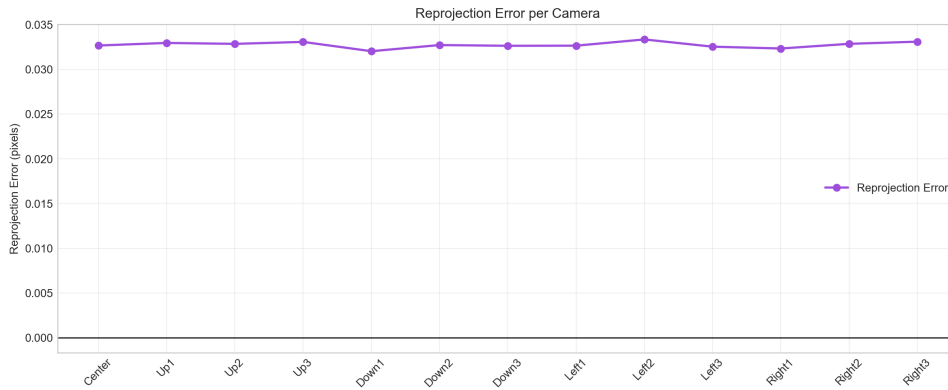


Figure 31: Reprojection error per camera of the initial calibration (Imageset2).

A.1.3. Imageset3

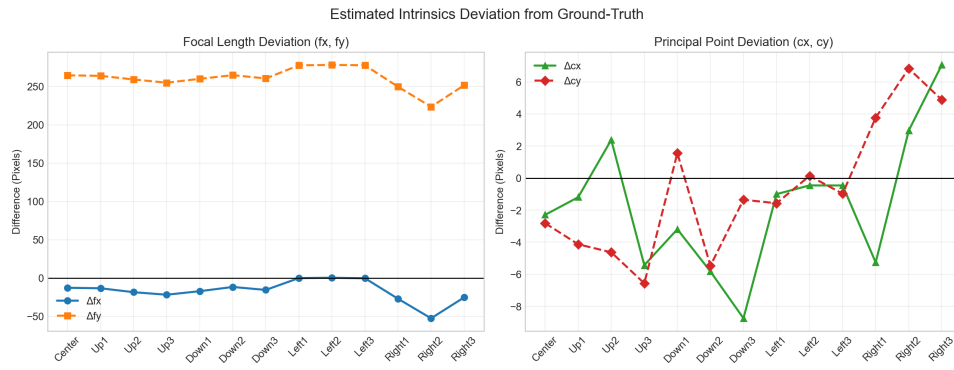


Figure 32: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset3).

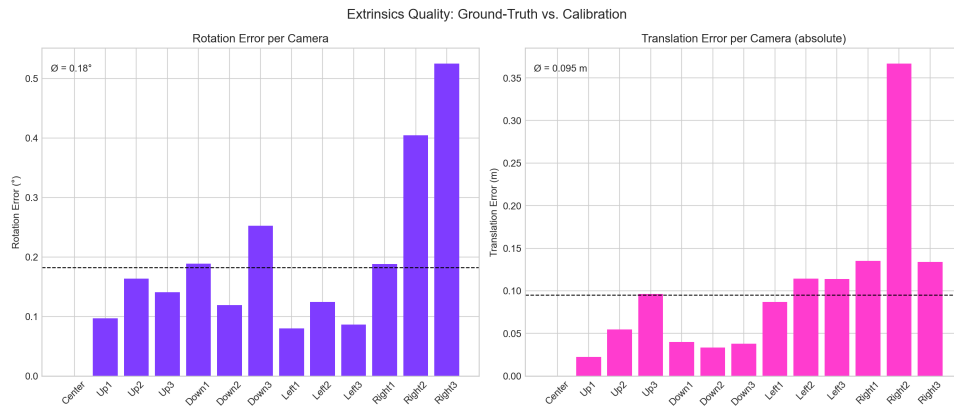


Figure 33: Extrinsic parameter errors of the initial calibration (Imageset3).

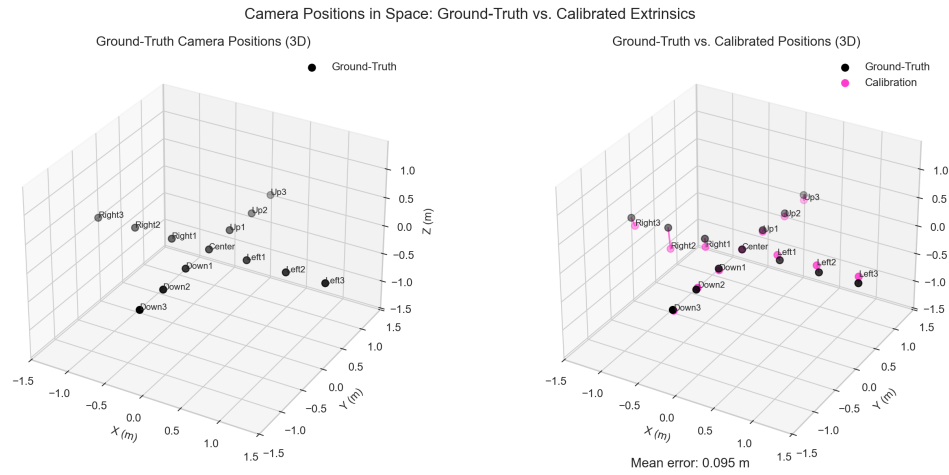


Figure 34: Translation parameter deviations from ground truth and the initial calibration (imageset3).

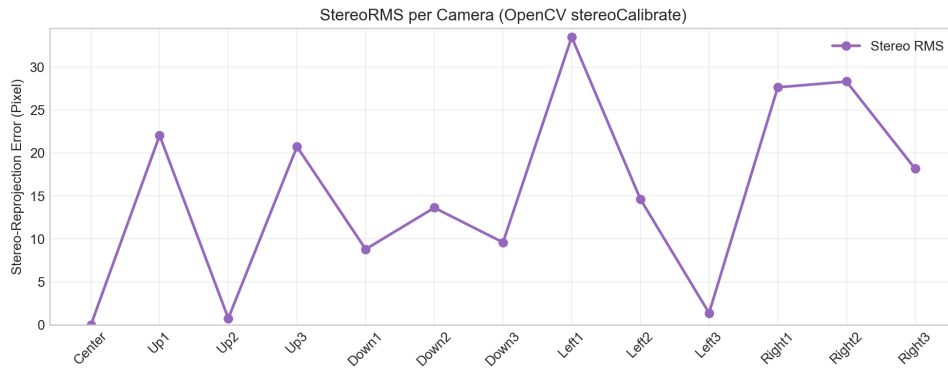


Figure 35: Stereo RMS error of the initial calibration (Imageset3).

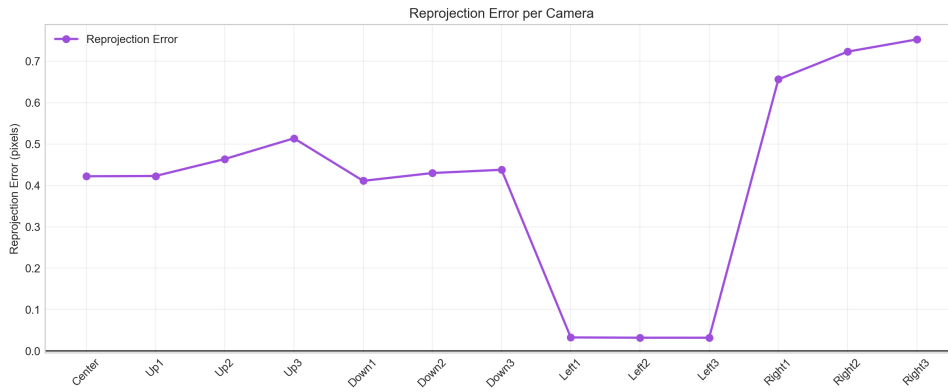


Figure 36: Reprojection error per camera of the initial calibration (Imageset3).

A.1.4. Imageset4

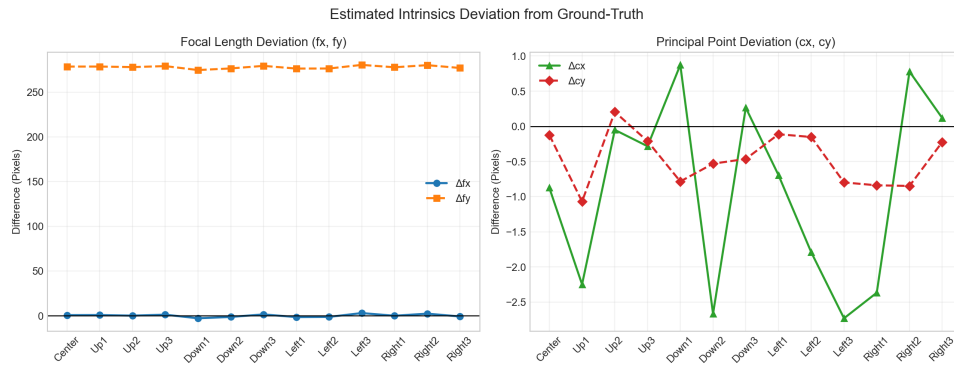


Figure 37: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset4).

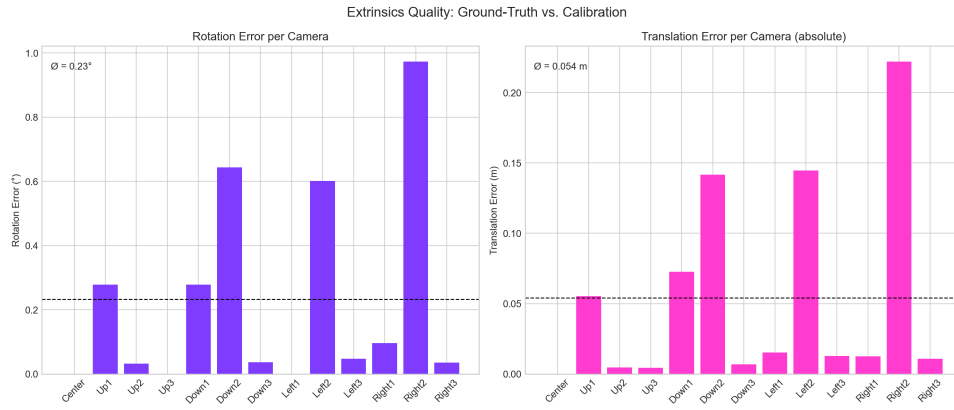


Figure 38: Extrinsic parameter errors of the initial calibration (Imageset4).

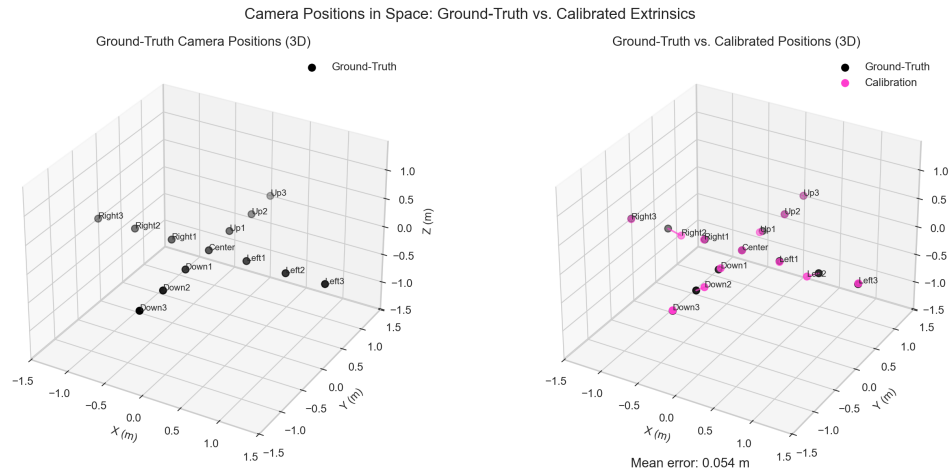


Figure 39: Translation parameter deviations from ground truth and the initial calibration (Imageset4).

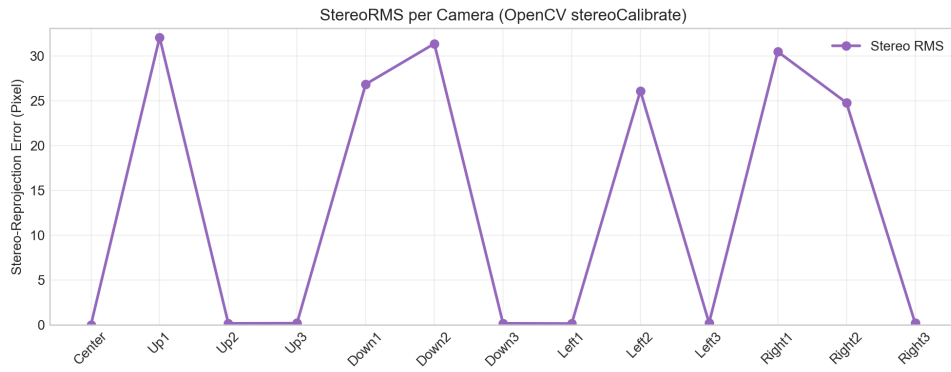


Figure 40: Stereo RMS error of the initial calibration (Imageset4).

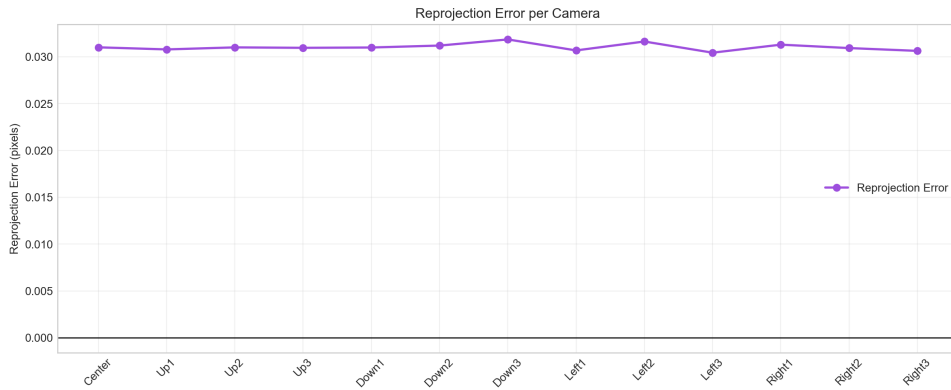


Figure 41: Reprojection error per camera of the initial calibration (Imageset4).

A.1.5. Imageset6

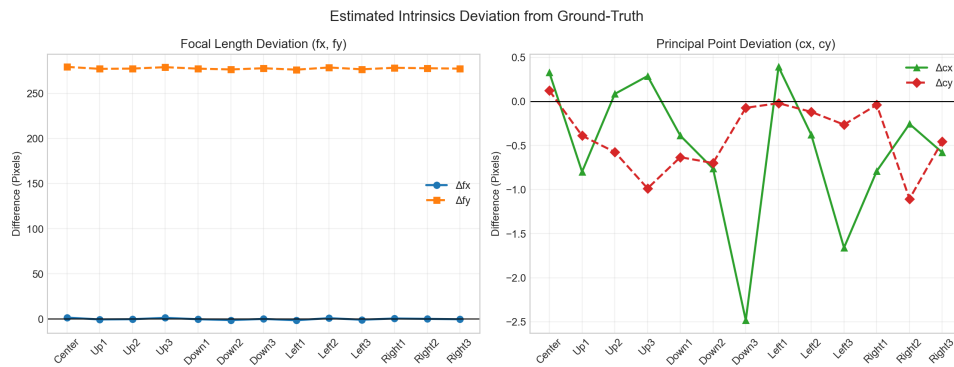


Figure 42: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset6).

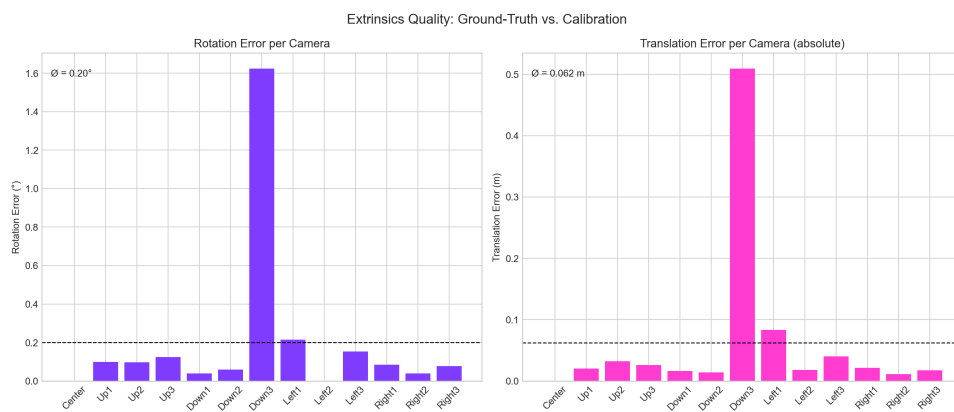


Figure 43: Extrinsic parameter errors of the initial calibration (Imageset6).

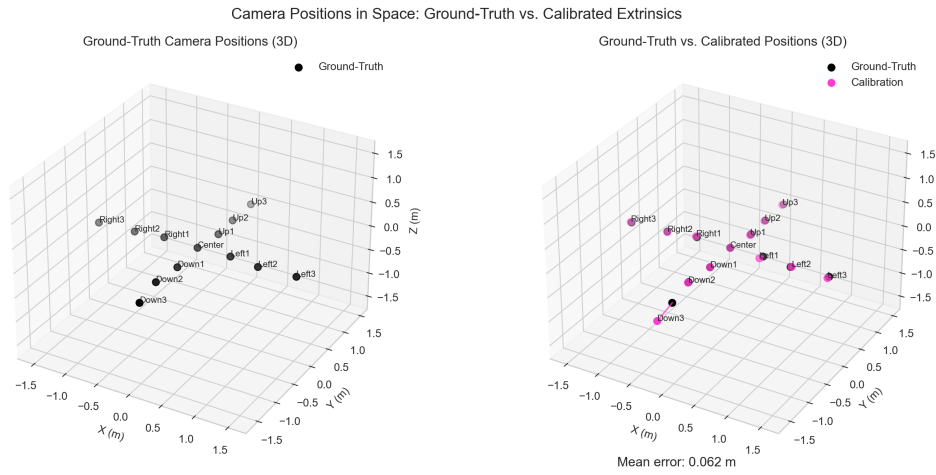


Figure 44: Translation parameter deviations from ground truth and the initial calibration (Imageset6).

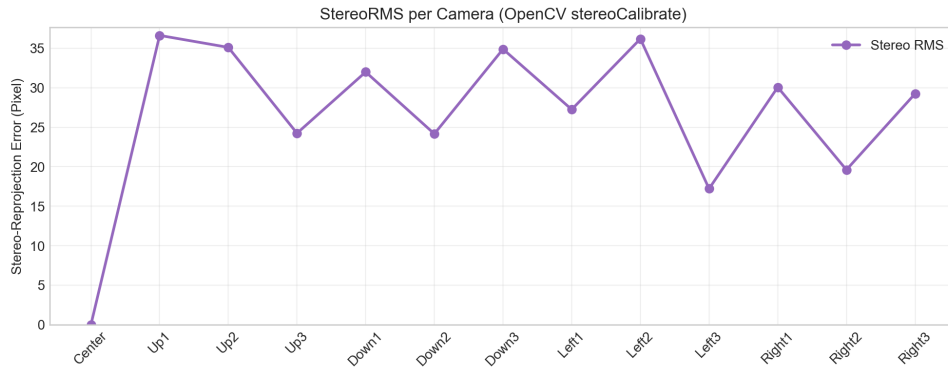


Figure 45: Stereo RMS error of the initial calibration (Imageset6).

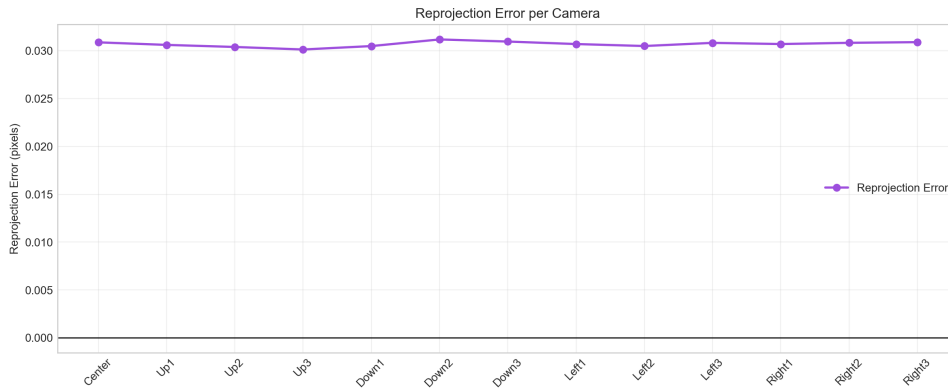


Figure 46: Reprojection error per camera of the initial calibration (Imageset6).

A.1.6. Imageset7

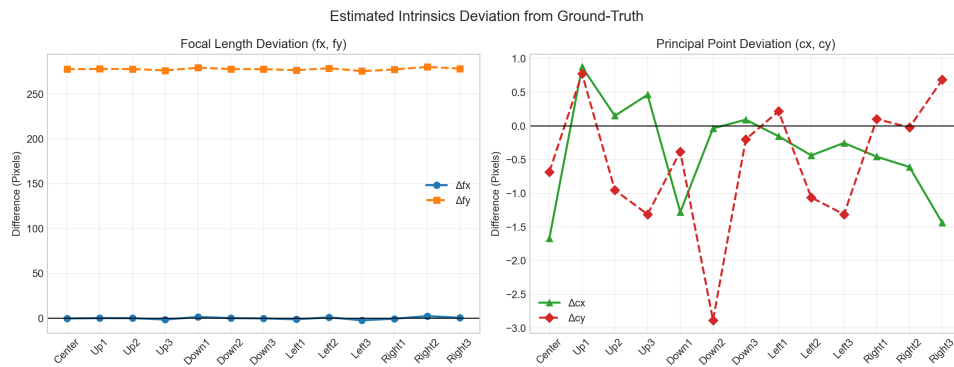


Figure 47: Deviations of intrinsic parameters from ground truth for the initial calibration (Imageset7).

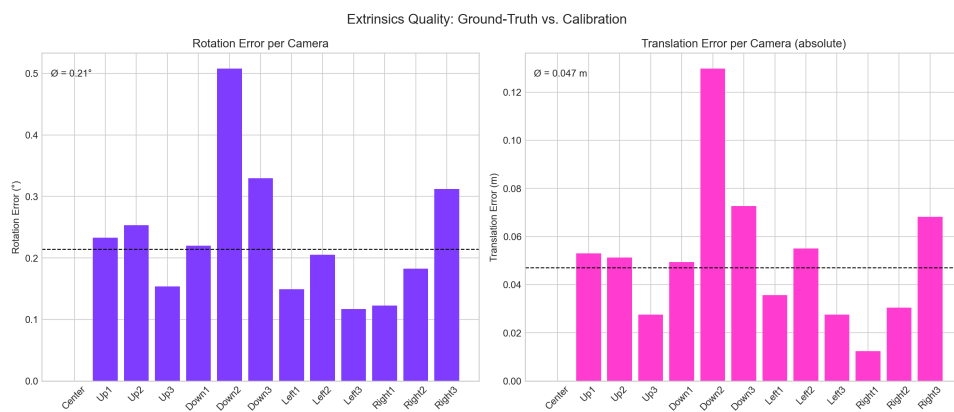


Figure 48: Extrinsic parameter errors of the initial calibration (Imageset7).

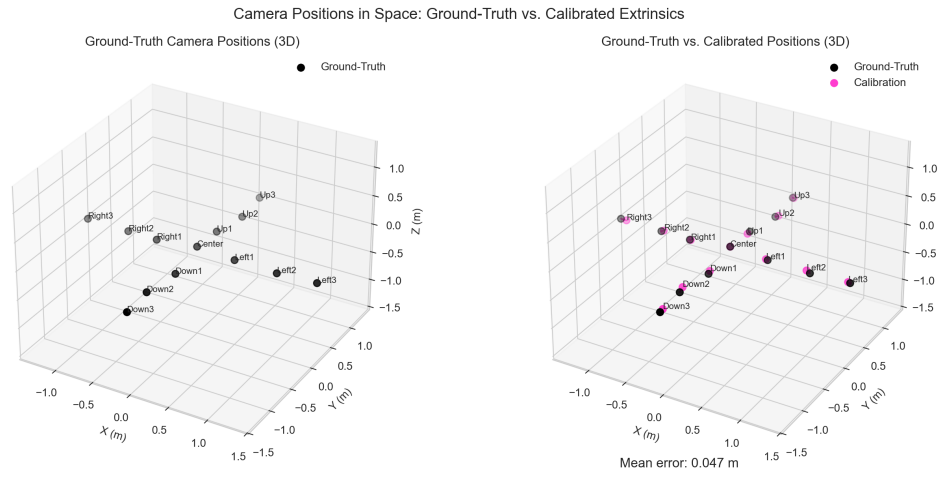


Figure 49: Translation parameter deviations from ground truth and the initial calibration (Imageset7).

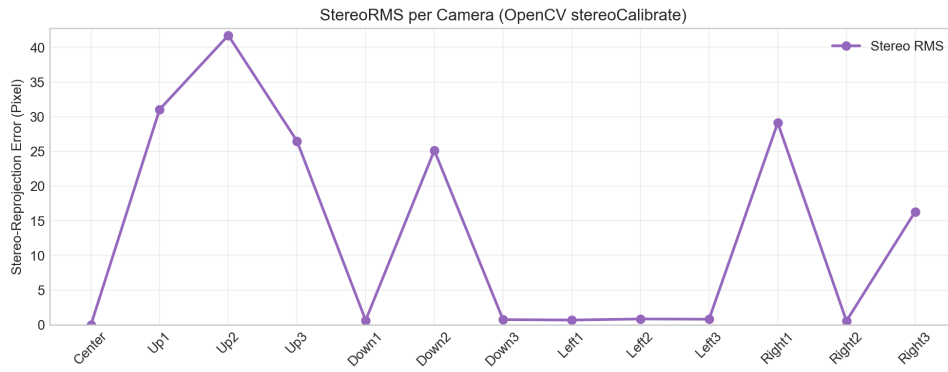


Figure 50: Stereo RMS error of the initial calibration (Imageset7).

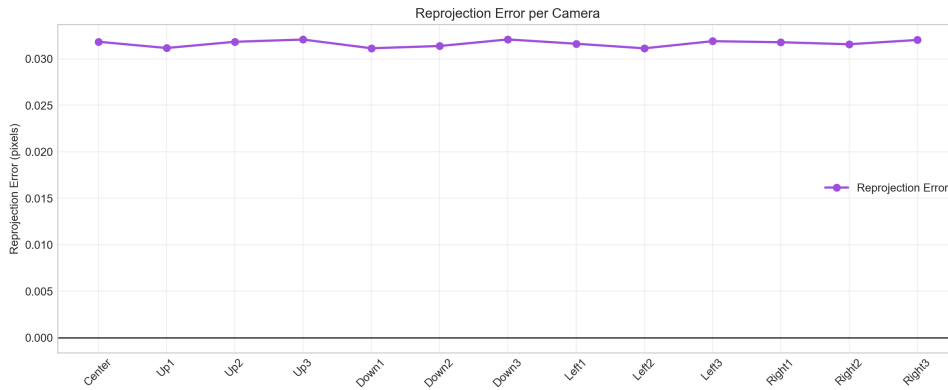


Figure 51: Reprojection error per camera of the initial calibration (Imageset7).

A.2. Blender Scenes

The Blender scenes used to generate the synthetic image data for the evaluations. The first plot shows the scene used for the initial target-based calibration, while the second plot illustrates the scene employed for the LiFCal-based recalibration.

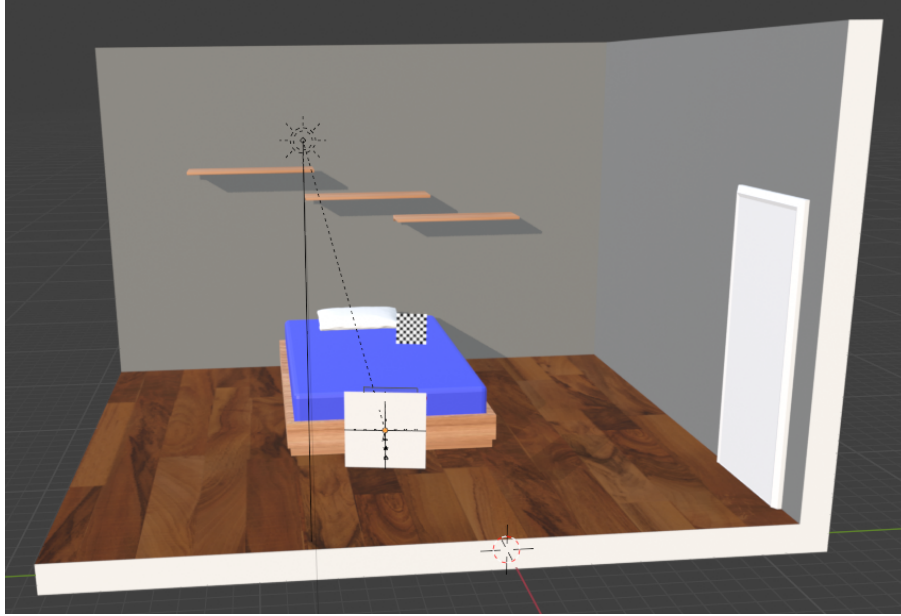


Figure 52: Blender scene for initial target-based calibration with checkerboard.



Figure 53: Blender scene for LiFCal-based recalibration without checkerboard.

Glossary

Light field Information about light rays in a scene, not only their brightness but also their direction. This enables refocusing and better depth cues.

Light field camera system A system built to capture a light field, either with a camera array or with a micro-lens array (MLA).

Multi-camera array Several cameras arranged in a fixed geometry to capture the same scene from different viewpoints.

Multi-camera light field system A light field setup based on a camera array (in this thesis, a cross-shaped layout).

Micro-lens array (MLA) A grid of tiny lenses in front of one sensor that captures angular information in a compact light field camera.

Pinhole camera model A simple camera model that describes how 3D points are projected onto the 2D image plane.

Intrinsic parameters Parameters inside the camera model, mainly focal lengths (f_x, f_y) and principal point (c_x, c_y) (in pixels).

Extrinsic parameters The camera pose in space, usually rotation R and translation t .

Camera matrix (K) Matrix that bundles the intrinsic parameters.

Principal point (c_x, c_y) Where the optical axis hits the image sensor (in pixel coordinates).

Radial distortion Lens distortion that bends straight lines, stronger toward the image borders (barrel/pincushion effect).

Tangential distortion Distortion caused by lens-sensor misalignment, modeled with p_1 and p_2 .

Calibration target A physical pattern with known geometry (e.g., a checkerboard) used to calibrate cameras.

Checkerboard calibration A common target-based calibration where checkerboard corners are detected in many images.

Target-free calibration Calibration without a physical pattern, based on scene structure, features, and usually camera motion.

Data-driven calibration It refers to an approach in which decisions and evaluations are based on measured data rather than predefined assumptions. In the context of this thesis, data-driven means that calibration quality is assessed using observed image data and error metrics, which then guide automatic decisions within the system.

LiFCal A target-free calibration method (originally for MLA light field cameras) used here as a baseline idea for automatic recalibration.

Feature detection Finding distinctive points in an image that can be matched across views (needed for many target-free methods).

Depth map An image where each pixel stores depth. In this thesis, depth maps come from Blender and support LiFCal-style processing.

Reprojection error Average pixel distance between detected points and the points projected by the estimated camera model. Lower is better.

Root mean square (RMS) error A single aggregated error value over many points. In OpenCV stereo calibration it reflects joint consistency between two views.

Stereo calibration (`stereoCalibrate`) OpenCV routine that estimates the relative pose between two cameras and returns an RMS error for that pair.

Bundle adjustment (BA) A global refinement step that improves parameters by minimizing reprojection errors across many observations.

Baseline calibration The stored reference calibration (from the initial checkerboard run) used to judge later recalibration attempts.

Recalibration Repeating calibration after the initial setup to update parameters when the system changes.

Calibration drift Calibration slowly becomes worse over time (e.g., small mechanical shifts), which motivates periodic recalibration.

Health score A single score in $[0, 100]$ used here to summarize calibration quality and decide whether a recalibration is accepted.

Ground truth Reference parameters taken directly from Blender, used to evaluate calibration accuracy in a controlled setting.

Synchronization How well cameras capture images at the same time. Timing offsets can harm multi-camera consistency.

Texture / scene structure Visible detail in a scene (edges, patterns, depth variation) that makes target-free calibration more reliable.

References

- [1] Yuriy Anisimov, Gerd Reis, and Didier Stricker. Calibration and auto-refinement for light field cameras. In *Proceedings of the 29th International Conference in Central Europe on Computer Graphics, Visualization and Computer Vision (WSCG)*, pages 253–261, Pilsen, Czech Republic, 2021. Vaclav Skala - UNION Agency. ISBN 978-80-86943-34-3. URL https://www.dfki.de/fileadmin/user_upload/import/11659_wscg-j53.pdf.
- [2] CamCalib. The reprojection error. <https://www.camcalib.io/post/what-is-the-reprojection-error>, 2022. CamCalib blog, accessed 2026-01-21.
- [3] Blender Online Community. *Blender - a 3D modelling and rendering package*. Blender Foundation, Stichting Blender Foundation, Amsterdam, 2018. URL <http://www.blender.org>.
- [4] Daniel Cremers. Bundle adjustment & nonlinear optimization. https://cvg.cit.tum.de/_media/teaching/ss2019/mvg2019/material/multiviewgeometry7.pdf, 2019. Vorlesungsunterlage, Technische Universität München.
- [5] Elijs Dima, Mårten Sjöström, and Roger Olsson. Assessment of multi-camera calibration algorithms for two-dimensional camera arrays relative to ground truth position and direction. In *2016 3DTV-Conference: The True Vision - Capture, Transmission and Display of 3D Video (3DTV-CON)*, pages 1–4, 2016. doi: 10.1109/3DTV.2016.7548887. URL <https://ieeexplore.ieee.org/document/7548887>.
- [6] Elsevier. Radial distortion. <https://www.sciencedirect.com/topics/computer-science/radial-distortion>, 2022. Definition from ScienceDirect Topics: Computer Science.
- [7] Elsevier. Lens distortion. <https://www.sciencedirect.com/topics/engineering/lens-distortion>, 2024.
- [8] Aymeric Fleith, Doaa Ahmed, Daniel Cremers, and Niclas Zeller. Lifcal: Online light field camera calibration via bundle adjustment. <https://arxiv.org/abs/2408.11682>, 2024.
- [9] Pengxiang Ge, Huanqing Wang, Yonghong Wang, and Biao Wang. Calibration of ring multicamera system with transparent target for panoramic measurement. *IEEE Sensors Journal*, 22(23):23154–23164, 2022. doi: 10.1109/JSEN.2022.3214080. URL <https://ieeexplore.ieee.org/document/9923591>.
- [10] German Aerospace Center (DLR) – Institute of Robotics and Mechatronics. Light field cameras. DLR Research Expertise: 3D Perception, 2026. URL <https://www.dlr.de/en/rm/research/expertise/3d-perception/light-field-cameras>. Accessed January 24, 2026.
- [11] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, Cambridge, UK, 2nd edition, 2003. ISBN 0521540518. URL http://www.r-5.org/files/books/computers/algo-list/image-processing/vision/Richard_Hartley_Andrew_Zisserman-Multiple_View_Geometry_in_Computer_Vision-EN.pdf.

- [12] Jiachun Huang, Shaoli Liu, Jianhua Liu, and Zehua Jian. Camera calibration results: Reprojection error and relative positions. *Scientific Reports*, 2016. URL https://www.researchgate.net/figure/Camera-calibration-results-a-Reprojection-error-and-b-relative-positions-of-the_fig9_301715405. In Fig. 5.
- [13] IEEE. Ieee recommended practice for software requirements specifications. *IEEE Std 830-1998*, pages 1–40, 1998. doi: 10.1109/IEEESTD.1998.88286.
- [14] Mark Hedley Jones. Calibration checkerboard collection. Online project page, 2025. URL <https://markhedleyjones.com/projects/calibration-checkerboard-collection>.
- [15] Dhiraj Kumar. What is bundle adjustment? Baeldung on Computer Science, 2024. URL <https://www.baeldung.com/cs/computer-vision-bundle-adjustment>. Last updated March 18, 2024.
- [16] Philip K. Lee and Anqi R. Ji. Comparison of lightfield reconstruction methods using undersampled data. Technical Report EE367 Winter 2017 Project Report, Stanford University, Department of Electrical Engineering, 2017. URL https://stanford.edu/class/ee367/Winter2017/lee_ji_ee367_win17_report.pdf.
- [17] Fei-Fei Li et al. Camera models. https://web.stanford.edu/class/cs231a/course_notes/01-camera-models.pdf, 2020.
- [18] Shih-Hao Lin, En-Tze Chen, Jia-Jun Xiao, and Ching-Yuan Chang. Stereo digital image correlation (3d-dic) for non-contact measurement and refinement of delta robot arm displacement and jerk. *The International Journal of Advanced Manufacturing Technology*, 133(5-6):2809–2822, July 2024. doi: 10.1007/s00170-024-13957-2. URL <https://doi.org/10.1007/s00170-024-13957-2>.
- [19] MathWorks. Camera calibration. <https://de.mathworks.com/help/vision/ug/camera-calibration.html>, 2024.
- [20] OpenCV Contributors. Camera calibration with opencv. https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html, 2024.
- [21] OpenCV Developers. Camera calibration and 3d reconstruction — opencv documentation. OpenCV 4.x Documentation, calib3d module, 2025. URL https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html.
- [22] rahadianrayhan. Living room 3d model. Free3D online 3D model, 2021. URL <https://free3d.com/3d-model/living-room-997877.html>. Blender, FBX, and OBJ formats, 20 035 vertices, 18 735 faces, 38 514 triangles.
- [23] Carlos Ricolfe-Viala and Antonio-José Sánchez-Salmerón. Lens distortion models evaluation. *Applied Optics*, 49(30):5914–5928, 2010. doi: 10.1364/AO.49.005914. URL https://www.researchgate.net/publication/47510646_Lens_distortion_models_evaluation.
- [24] satriacaster. Isometric room 3d model. Free3D online 3D model, 2021. URL <https://free3d.com/3d-model/isometric-room-525234.html>. Blender (.blend), textures and materials included.

- [25] Chaehyeon Song, Dongjae Lee, Jongwoo Lim, and Ayoung Kim. Camera calibration via circular patterns: A comprehensive framework with measurement uncertainty and unbiased projection model. *arXiv preprint arXiv:2506.16842*, 2025. URL <https://arxiv.org/abs/2506.16842>.
- [26] Stanford Computer Graphics Laboratory. The stanford multi-camera array. Project page, Stanford University, 2011. URL <https://graphics.stanford.edu/projects/array/>.
- [27] Bill Triggs, Philip F. McLauchlan, Richard I. Hartley, and Andrew W. Fitzgibbon. Bundle adjustment – a modern synthesis. In Bill Triggs, Andrew Zisserman, and Richard Szeliski, editors, *Vision Algorithms: Theory and Practice*, volume 1883 of *Lecture Notes in Computer Science*, pages 298–372, London, UK, 2000. Springer-Verlag. doi: 10.1007/3-540-44480-7_21. URL <https://www.cs.jhu.edu/~misha/ReadingSeminar/Papers/Triggs00.pdf>.
- [28] Zhaomin Xie, Mengxiao Zhao, and Yujia He. Research on depth information acquisition of array camera based on depth map fusion. In *2023 5th International Conference on Frontiers Technology of Information and Computer (ICFTIC)*, pages 759–762, 2023. doi: 10.1109/ICFTIC59930.2023.10456113. URL <https://ieeexplore.ieee.org/document/10456113>.
- [29] Yichao Xu, Kazuki Maeno, Hajime Nagahara, and Rin ichiro Taniguchi. Camera array calibration for light field acquisition. *Frontiers of Computer Science*, 9(5): 691–702, 2015. doi: 10.1007/s11704-015-4237-4. URL <https://link.springer.com/article/10.1007/s11704-015-4237-4>.
- [30] Jungang Yang, Chao Xiao, Pu Wang, Yougan Luo, and Chengjin An. Camera array calibration using a simple checkerboard pattern. In *2019 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, pages 1–5, 2019. doi: 10.1109/ICSIDP47821.2019.9172948. URL <https://ieeexplore.ieee.org/document/9172948>.
- [31] Yaning Zhang, Yingqian Wang, Tianhao Wu, Jungang Yang, and Wei An. Fixed relative pose prior for camera array self-calibration. *IEEE Transactions on Circuits and Systems for Video Technology*, 35(1):981–985, 2025. doi: 10.1109/TCSVT.2024.3450706. URL <https://ieeexplore.ieee.org/document/10649654>.
- [32] Z. Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000. doi: 10.1109/34.888718. URL <https://ieeexplore.ieee.org/document/888718>.
- [33] Feng Zhou, Jing Li, Yanran Dai, Lichen Liu, Haidong Qin, Yuqi Jiang, Shikuan Hong, Bo Zhao, and Tao Yang. Time-series fusion-based multicamera self-calibration for free-view video generation in low-texture sports scene. *IEEE Sensors Journal*, 23(3):2956–2969, 2023. doi: 10.1109/JSEN.2022.3230792. URL <https://ieeexplore.ieee.org/document/9998530>.

Acknowledgment